

Prifysgol Wreccsam Wrexham University

Faculty of Arts, Computing and Engineering

Analysis of Recent Ransomware Attacks Using Cyber Incident Data

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Submitted by: Pavan Reddy Pagilla

Supervisor: Dr Colin Kuka

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Acknowledgement

This dissertation made limited use of artificial intelligence (AI) tools, including ChatGPT and Google Colab, to support code debugging, language refinement, and structural clarity. All core aspects of the research including problem formulation, data analysis, methodology design, implementation, and interpretation of results—were carried out independently by the author. AI tools did not contribute to the generation of original research findings. All outputs were critically reviewed, validated, and edited to ensure accuracy, originality, and compliance with academic integrity standards.

Declaration of Originality

I, Pavan Reddy Pagilla, hereby declare that this dissertation, submitted in partial fulfilment of the requirements for the degree of MSc Cyber Security with Advanced Practice at Wrexham University, is entirely my own work. All sources consulted have been duly acknowledged and referenced in accordance with the University's academic integrity guidelines. This work has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution.

Signature:

Name: Pavan Reddy Pagilla

Date: 23rd April 2026

Abstract:

Ransomware is now one of the most harmful and destructive forms of cybercrime that has impacted organisations, governments and critical infrastructure globally. The dissertation examines recent ransomware attacks using the EURepoC Dyadic Cyber Incident Dataset (2025), a publicly accessible dataset containing structured cyber incident data, such as those involving state-attributed and non-state actors. Three supervised machine learning models were used: Logistic Regression, Random Forest, and XGBoost. Ransomware incidents were analysed using these models based on characteristics such as attack severity, disruption type, attribution, and target region. Prior to the development of the models, exploratory data analysis (EDA) was conducted to identify temporal trends, region, and severity of the incident. Findings indicate that ransomware activity increased substantially after 2015. Ransomware incidents were more severe than other cyber incidents, with high effect sizes (Cohen's $d = 1.23$ to 1.31 , $p < 0.001$). The regional analysis showed that the regions with the highest concentration of ransomware attacks are in North America and NATO-aligned areas, indicating a clear geopolitical targeting pattern.

XGBoost scored the highest in terms of model performance with test AUC = 0.9593, and with Random Forest closely behind (AUC = 0.9585). The Logistic Regression baseline achieved acceptable performance (AUC = 0.9200). The analysis of feature importance and SHAP revealed that the most significant factors in predicting ransomware incidents are attribution identity, the severity of incidents, and geopolitical relations between nations. Overall, the results indicate that geopolitical and operational incident-level can be effectively used to classify ransomware incidents. The outcomes can be applied to cybersecurity threat intelligence, allowing analysts to prioritise high-risk events and make more informed decisions. This work can be extended in the future using time-series forecasting and multi-class attack classification.

Keywords: Ransomware, Cyber Incident Data, Machine Learning, XGBoost, SHAP, Threat Intelligence

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List of Abbreviations

Abbreviation	Full Form	Abbreviation	Full Form
CSIS	Center for Strategic and International Studies	NATO	North Atlantic Treaty Organisation
CTI	Cyber Threat Intelligence	NLP	Natural Language Processing
DBIR	Data Breach Investigations Report	PR-AUC	Precision–Recall Area Under the Curve
DPRK	Democratic People’s Republic of Korea	RaaS	Ransomware-as-a-Service
EDA	Exploratory Data Analysis	ROC	Receiver Operating Characteristic
ENISA	European Union Agency for Cybersecurity	SHAP	Shapley Additive Explanations
EURepoC	European Repository of Cyber Incidents	SIEM	Security Information and Event Management
GridSearchCV	Grid Search Cross-Validation	SMOTE	Synthetic Minority Oversampling Technique
MENA	Middle East and North Africa	SOC	Security Operations Centre
ML	Machine Learning	XGBoost	Extreme Gradient Boosting

Chapter 1: Introduction

1.1 Background

Ransomware is malware that encrypts or locks a victim's data and requests money to unlock it. As the years progressed, it has become a major global issue that has impacted on governments, businesses, hospitals and critical infrastructure. Ransomware has impacts beyond financial loss. These include service disruption, loss of data permanently, loss of reputation, and in severe instances, jeopardising public safety. One of the most notable examples is the 2017 WannaCry attack on the UK National Health Service which led to the cancellation of approximately 19,000 appointments and incurred upfront expenses over 92 million pounds [8]. This demonstrates that ransomware can have real-life consequences beyond computer networks. The governments of the United States, the United Kingdom, and Australia claimed that the malware was the work of the North Korean Lazarus Group and used the Eternal Blue vulnerability to propagate across unpatched Windows systems around the world at a record pace. The malware spread to more than 200,000 systems across 150 countries in a matter of hours, and it affected the activities of hospitals, telecommunication systems, and manufacturing plants. The event proved that ransomware is not a mere criminal nuisance but a strategic threat that has geopolitical roots and effects that go further to the health of the population and national security. One reason ransomware is rapidly spreading is due to the emergence of Ransomware-as-a-Service (RaaS). In this scheme, criminal gangs sell off-the-shelf ransomware to other criminals at a commission on the ransom. This has not only reduced the technical expertise to conduct attacks but it has also enhanced their frequency across the globe. According to [24], the global ransomware epidemic can be seen as one of the hallmarks of the cybercrime of the twenty-first century where the ecosystem has stopped being opportunistic and is characterized by the targeted and high-value operations against critical infrastructure and enterprise victims. The costs are huge in financial terms: ransom payments are just a portion of the overall expenses of incidents when operational disruption, recovery costs, legal liability, and reputational losses are also considered [3]. Ransomware is also associated with geopolitical factors. [23] demonstrate that geopolitical tensions are associated with the rise of ransomware operations, and it becomes difficult to differentiate between financially-oriented cybercrime and state-sponsored cyber operations. Ransomware is not only a commercial security issue, but a

national and international security challenge because some of the attacks are linked to state sponsored or state-tolerated organizations [11]. The extent of the issue is echoed in the economic statistics. [17] Recorded an over twofold rise in ransomware assaults on healthcare organisations in the US during 2016-2021, and the cases of data exfiltration along with encryption are on the rise. The bigger picture of the critical national infrastructure risk also justifies the strategic framing that is embraced in this dissertation. In May 2021, the Colonial Pipeline ransomware attack, which was carried out by the Darkside RaaS group, had to shut a six-day-long larger refined-products pipeline in the United States and illustrated that ransomware can be used to generate a real systemic risk to the national economic infrastructure [19]. One month later, the JBS Foods attack shook the capacity of processing meat all over the world. The Kaseya VSA attack was an attack in July 2021 that ransomware was installed with a software update by a managed service provider, and it impacted more than 1,500 downstream organisations at the same time. All these events demonstrate that ransomware has outgrown traditional cybercrime to become a threat to the economy and supply chain integrity. Sectoral concentration of ransomware recorded in the literature and verified in EDA of this study (Section 5.1.5) indicates that ransomware operators have intentional targeting decisions. The targeted healthcare organisations are due to the fact that patient care requires real-time access to medical records and imaging systems, which puts a lot of pressure on organisations to get back to their feet. Government agencies are targeted since they store high amounts of sensitive data about citizens and are under pressure to solve incidents politically. Critical infrastructure operators are the targets because of the societal effect of prolonged outages, whether they are power, water, or transport, which offer leverage that cannot be equalled by commercial interests that are purely commercial in nature. These targeting patterns are listed in the receiver subcategory column of the EURPoC dataset and they validate the notion that ransomware victimisation is not random but follows some predictable patterns due to the assessment of vulnerability and willingness to pay by attackers. The growing complexity of ransomware activities is also manifested in the model of the double extortion that has been the leading mode of operation of large RaaS groups since around 2019. Under this model attackers not only encrypt data belonging to victims, but also steal sensitive information prior to triggering encryption and threatening to publish stolen data on special leak sites in case the ransom is not paid. This puts in place a two-fold coercion effect: although the victim may have working backups and may be able to re-establish operation without paying to decrypt, they may risk sensitive data leakage with ensuing

regulation fines, competitive loss and reputation. The key difference of the double extortion model is that the economy of ransomware response is altered: organisations can no longer consider ransomware an availability issue that can be resolved with the help of backup restoration, but also a confidentiality aspect of the attack.

1.2 Research Context

The majority of research done on ransomware currently concentrates on technical detection technologies, including malware analysis, file behaviour, and network traffic monitoring. Although these methods are relevant in detecting attacks on individual systems, they fail to capture the bigger picture of cyber-attacks like who was attacked, where and the extent of the attack. There is an increased necessity to research ransomware with the help of structured cyber incident data that encompasses information about attack characteristics, the affected area, and attribution.

This is precisely the kind of structured, geopolitically contextualised data that is offered by the EURepoC Dyadic Cyber Incident Dataset. The dataset, which is the compilation of a European based research consortium, includes 4,296 dyadic records, including the initiator-receiver relationship between countries, the type of attack, the scores of severity, the identities of the attributions, and the estimations of the economic effects between 2000 and 2024. Nevertheless, it has not been widely used in machine learning classification of ransomware incidents which is the gap that this study fills. Using a supervised machine learning pipeline on this data, this paper contributes an empirical study and a reproducible methodology structure to this area.

Cyber Threat Intelligence (CTI) provides the theoretical framing within which this research is situated. CTI refers to the process of collecting, analysing, and applying knowledge about cyber threats to support informed defensive decision-making at tactical, operational, and strategic levels. At the strategic level, CTI is concerned with understanding threat actor motivations, targeting patterns, and geopolitical context, precisely the type of analysis that structured incident databases such as EURepoC are designed to support. The geopolitical dimension of ransomware, wherein financially motivated criminal groups operate with varying degrees of state tolerance and target organisations in adversary-aligned regions, means that CTI frameworks are increasingly relevant to ransomware analysis beyond the purely technical domain. This study contributes to the strategic CTI literature by demonstrating that machine learning on geopolitically contextualised incident data can

produce interpretable, actionable threat characterisation. One should not confuse three closely related yet distinct analytical objectives of the ransomware machine learning literature: detection (identifying ransomware during an attack in real-time), classification (identifying whether a completed incident was ransomware or not based on structured records), and characterisation (describing the patterns that differentiate ransomware and other types of incidents). Instead of the real-time detection, the classification and characterisation are examined in this dissertation. This defines the points of the relevant comparison in the literature, the most suitable metrics of performance, and the main practical users, which are threat intelligence analysts and strategic security functions as opposed to endpoint security engineers.

1.3 Research Aim

This study examines ransomware attacks based on the EURPoC data and implement machine learning to see which factors best predict the occurrence of ransomware.

1.4 Research Objectives

1. To perform an exploratory data analysis (EDA) to determine patterns and trends in ransomware cases.
2. To develop and test three machine learning models (Logistic Regression, Random Forest, and XGBoost) to classify ransomware incidents.
3. To measure model performance using AUC-ROC, precision, recall, and F1-score.
4. To identify the most important features using feature importance and SHAP analysis.
5. To connect the findings with other studies in cybersecurity.

1.5 Research Hypotheses

H1: Ransomware cases are far more severe as compared to non-ransomware cyber cases.

H2: North American and NATO-aligned regions are more likely to experience ransomware attacks than other regions globally.

1.6 Significance of the study

The work also adds to the literature on cybersecurity by utilizing machine learning on structured data on cyber incidents and not on technical characteristics of malware. It shows that ransomware incidents can be predicted using geopolitical and operational factors, which has practical implications to threat intelligence and incident prioritisation. Theoretically, it generalizes machine learning usage on organized records of geopolitical cyber incidents, a field that has been scantily methodologically investigated. In a practical sense, the research findings provide cybersecurity experts with empirically confirmed information that the severity of the incident score, attribution identity, and geopolitical structure are the most discriminative factors in organized incident databases of ransomware activity. This directly affects threat intelligence prioritisation, security operations centre triage process and automated incident analysis pipeline design.

This study contributes to the literature by demonstrating that openly accessible geopolitical incident data, which is managed by academic research consortia, as opposed to commercial threat intelligence vendors, can be used to classify the machine learning, at a performance level, on par with that of proprietary technical data. It has consequences on the availability of cybersecurity analytical research: when datasets with strong classification performance are supported by openly licensed datasets, like EURepoC, the analytical frameworks constructed here can be generalized by scholars and professionals without having to spend on commercial data subscriptions. The research also adds a reproducible method pipeline, which involves fivefold GridSearchCV optimisation, stratified class imbalance control, multi-metric evaluation and dummy baselines, SHAP interpretability and systematic feature ablation that can be generalised to any organised cyber incident dataset with binary or multi-class attack type outcomes.

1.7 Scope and Limitations

The research is founded on EURepoC Dyadic Cyber Incident Dataset (2025) and aims at binary classifying ransomware and non-ransomware incidents. No primary data gathering was developed and no human subjects were used. Some of the limitations are that it will depend on publicly reported data, which may not be accurate in attribution, and historical data that may not be a true reflection of the current trends. The dataset includes the period between January 2000 and December 2024, which includes the pre-ransomware baseline

period and the current ransomware epidemic and makes it possible to conduct the structural break analysis, one of the most important EDA results.

1.8 Dissertation Structure

Chapter 2 presents the problem statement. Chapter 3 discusses the available literature. The methodology is given in chapter 4. Chapter 5 shows the results and analysis. The findings are discussed in chapter 6. Chapter 7 presents conclusions. In Chapter 8 the limitations and future research directions are discussed. Appendix A documents the EURepoC column schema. Appendix B provides the Google Colab link to the Python analysis notebook. Appendix C lists hyperparameter search grids. Appendix D contains full classification reports. Appendix E presents Pearson correlations. Appendix F contains all figures and visualisations. Appendix G presents summary tables.

Chapter 2: Problem Statement

2.1 Research Problem

Ransomware is a significant cybersecurity threat that leads to a large-scale loss of money and disruption of operations. The majority of the research that is available is concerned with the technical method of malware behaviour, file analysis, and network traffic detection. Although these techniques help in identifying attacks on individual systems, they fail to reveal the larger trends on ransomware attacks like where the attacks occur, the severity of the attacks and the perpetrators.

Little current research uses machine learning on structured cyber incident data incorporating geopolitical, regional, and operational data. EURepoC data provides a good chance to address this gap. This paper seeks to fill the gap by examining ransomware through incident-level information and predictive modelling. The fact that the EURepoC data has rich metadata with severity scores, attribution identities, dyadic relationships between countries, and time trends has not been effectively utilized in machine learning classification, and there is a huge gap between the potential and actual use of the data in the machine learning cybersecurity literature.

Another aspect of the issue has to do with attribution dependence. Attribution In the realm of cyber incident classification, attribution is the official process of identifying the threat actor who perpetrated an attack, and such an operation must involve intelligence tradecraft, technical forensics, and others, usually with the exchange of diplomatic information. Attribution is by its very nature uncertainties, subject to dispute, and in most instances never conclusively determined. The EURepoC dataset documents attribution of only formally attributed incidents by governments, security companies or courts, forming a skewed sample, which is biased towards incidents involving identifiable actor groups. An attribution model trained on attribution features of this population might not reliably generalise to most ransomware attacks by less well-known criminal groups that are not formally attributed. The ablation of features, as in this paper, is a necessary way of quantifying this generalisation risk to any plausible evaluation of readiness to deploy models at any given time. Also, most of the published ransomware ML researches do not have principled class imbalance treatment, dummy classifier baselines, and Precision-Recall AUC reporting, and so their performance

results are hard to place in context. In this study, all three deficiencies are tackled at the same time.

Another related issue is that most similar published work lacks the use of dummy baseline comparisons. Model performance must be interpreted relative to baseline classifiers to ensure validity. But unless the performance is compared to that of a dummy classifier which never misses the majority class (which attains 87% accuracy on this dataset whilst identifying zero ransomware cases) the reader is unable to determine how much performance came as a result of genuine learned patterns and how much as a result of class distribution artefacts. This paper takes care of this by comparing all supervised models in a systematic way to the Most Frequent and the Stratified dummy classifier, and giving clean decomposition of the performance into learned signal and the baseline noise.

Feature ablation is required to evaluate model robustness under feature constraints study in most published ransomware ML studies has more than methodological implications. When practitioners read published research that reported high AUC scores, they are not able to tell whether the model would be useful or not when some features are not available in their particular operational situation. Taking into account a case in point, a model with AUC 0.96 when using the attribution data might not be of value to an organisation, which lacks access to real-time attribution intelligence. The value of the published models in practical deployment is unknown without explicit ablation experiments to remove the attribution features, and quantify the cost of the resulting performance. The gap is bridged by this research directly, through the systematic ablation experiments that quantify the cost of unavailability of features, and give the practitioners the information they require to make informed decisions in deploying the features.

2.2 Research Questions

1. Which are the major trends and tendencies in the occurrence of ransomware in the EURRepoC dataset?
2. To what extent are machine learning models able to classify incident-level features to identify ransomware incidents?
3. What characteristics are the most significant to predict ransomware activity?

4. What are the differences between ransomware and non-ransomware incidents in terms of severity and targeting by region?

2.3 Study contribution.

The paper is also relevant to cybersecurity since it utilizes machine learning on an organized body of incidents, which offers a novel viewpoint to the conventional malware-centered methods. It also determines the important operational and geopolitical variables associated with ransomware and provides viable information to threat intelligence and decision-making. The hybrid analytical model, which includes the EDA and ML elements, will be required to answer both the classification question (is a model reliable to detect ransomware?), and the characterisation question (what patterns are unique to ransomware and not other types of incidents?).

The integrated analytical model that involves both EDA and ML segments answers two analytically similar yet different questions. The classification question is as follows: could a machine learning model with structured incident metadata reliably detect ransomware incidents? The characterisation question is: what are the targeting, timing, severity and attribution patterns that make ransomware different as compared to other types of cyber incidents? The characteristics that are most suitable to classify ransomware are also the ones that most effectively distinguish it against other incidents, although they demand other methods of analysis. The performance of classification is measured using machine learning metrics like AUC-ROC and PR-AUC and geopolitical characterisation is measured using the exploratory data analysis and the hypothesis testing. The two elements are required to provide a rigorous answer to both questions, and so this study combines them into a single line of analysis, and not as different research.

The study time period, which is January 2000 to December 2024 is analytically important. It includes the pre-ransomware baseline phase (2000-2014) where ransomware comprised around 2.1% of reported attacks, and the current ransomware pandemic (2015-2024) when it reached around 16.3%. It is critical to include the pre-2015 period in the formation of the historical baseline against which the post-2015 escalation will be measured; otherwise, the analysis of the temporal trends would have no context. The fact that the dataset includes the latest ransomware activity up to December 2024 means that the latest ransomware wave, namely the attacks on the healthcare sector in 2022-2024, as well as the geopolitical

ransomware threats that were part of the Russian-Ukraine conflict, are reflected in the analytical population.

Chapter 3: Literature Review

3.1 Introduction

The chapter provides a review of the current literature on ransomware, machine learning in cybersecurity, and analyzing cyber incident data. The goal is to learn what is already known, what main themes are presented and what gaps this research fills. Three main areas are covered: the evolution of ransomware, empirical studies on ransomware incidents, and machine learning approaches used in cybersecurity.

3.2 Ransomware: Definition and Evolution

Ransomware is a type of malicious software that denies access to data or systems, typically by encryption, until a ransom is paid. First ransomware was straightforward and employed rudimentary encryption spread over physical media. The contemporary ransomware is much more sophisticated with powerful asymmetric encryption algorithms, data theft, and multiplestage attack plans [18]. [11] present a comprehensive forensic taxonomy of Windows-based ransomware attack chains, which consists of six consecutive stages of operation: initial access, execution, persistence, privilege escalation, encryption, and ransom demand, which are specifically aimed at maximising the severity of the incident and reducing the possibility of defensive intervention. One of the significant trends is double extortion, by which the attackers not only encrypt data but also threaten to publish it online. This adds pressure to the victims and renders the attacks more destructive. The rise of Ransomware-as-a-Service (RaaS) has made it easier for groups without high technical skills to carry out attacks, which has led to a large increase in incidents worldwide [11]. [24] records the development of professionalised criminal networks that have the features of legitimate companies, such as customer support services, negotiation departments, and media relations services. The development of ransomware delivery systems would be pertinent to the geopolitical contextualization of the data utilized in this research. The first versions of ransomware were based on mass emailing with little specificity, resulting in high-volume and low-severe cases. The shift to targeted operations, made possible by the RaaS model has fundamentally altered the risk profile. Modern ransomware authors engage in lengthy reconnaissance campaigns prior to instigating encryption, network topology, critical data repository mapping, and persistence via a variety of access vectors. It is this targeting logic which causes the severity

differentials and regional distributions of concentration that can be seen in the EURPoC dataset: the events that are reported to have occurred in a geopolitical focused database are not the mass-market criminal ransomware targeting individual consumers but rather the targeted, high-severity operations. The monetary properties of the RaaS model have certain implications on the features that are captured in the EURPoC dataset. RaaS affiliates normally collect 70-80% of the ransom payment, with the rest 20-30% distributed to the team of core developers. This revenue-sharing model places pressure on affiliates to ensure each operation is optimised and as a result, causes the targeting of high-value organisations in affluent countries. This financial damage concentration is partially empirically reflected in the economic impact value field of the EURPoC dataset, which has a high sparsity (76.4% missing). The severity metrics (weighted intensity, unweighted intensity, impact indicator score) reflect the dimensions of operational impacts that are maximised by RaaS operations specifically. However, much of the taxonomy literature, including [11], is based on Windows-platform case studies and may not fully capture the growing targeting of Linux-based infrastructure and cloud environments, a limitation worth acknowledging when generalising these frameworks to the broader threat landscape.

3.3 Growth and Impact of Ransomware

Studies have always indicated that cases of ransomware have escalated drastically over the past years. Several studies point to a sharp acceleration that started to occur around 2015-2016, which is linked to the maturation of cryptocurrency payment systems and the emergence of early RaaS platforms [13]. One of the most extensively researched ones is the WannaCry attack of 2017, which has interfered with the work of healthcare facilities worldwide, costing it billions of dollars [8].

Ransomware has significant financial and operational impact, as widely reported in prior studies. Expenses encompass ransom fees, recovery fees, downtime and reputation loss. All these effects make ransomware one of the most expensive types of cybercrime. [17] reported that ransomware attacks on US healthcare organisations have increased over twofold since 2016, and that they are now more likely to entail data exfiltration as well as encryption. [13] compare the nature of pre-pandemic and mid-pandemic ransomware attacks, reporting further acceleration during the COVID-19 era by increasing the surface area of remote work attacks.

[12] study 1,579 cyber incidents across various databases and discover that ransomware produces a total disproportionate cost in comparison with other types of incidents, despite adjusting organisational size and sector. This observation aligns with the severity differentials that were tested in the H1 hypothesis of this study and is independent evidence with another dataset that the severity attributes that made ransomware differentiate other incidents are real and not artefactual. [3] construct a multi-dimensional impact framework of cyber threats, which records that the public perception of the threat risk of ransomware significantly falls behind the real threat trend, indicating that some improved analytical utilities of the kind created in this paper may help resolve the information asymmetry within the threat intelligence ecosystem.

[20] use time-series modelling on ransomware incidents data and find recurring seasonal trends, which are indicative of the operational cycle of financially-driven organizations, and show significant predictive capacity in the short-term, as well as the near-term volume of incidents. This time prediction method is the complement to cross-sectional classification method used in this research: by using Roumani and Roumani to predict when an incident of ransomware will occur, the current research predicts which incidents are ransomware in relation to their operational nature. A deployed system that combines both of these approaches may offer a thorough threat characterisation both in the form of a temporal prediction as well as an incident classification.

[16] provide a critical methodological approach by coming up with statistical estimation approaches to counter systematic underreporting. In their analysis, they imply that the reported cases are less than 20 percent of the real ransomware activity, and smaller organisations are also significantly underrepresented. It is a valid point that any research including this one should take into account when using publicly-available databases of incidents. More broadly, much of the growth and impact literature relies on industry reports or healthcare-specific databases, which may introduce sector bias and limit the generalisability of findings across the full range of targeted organisations.

3.4 Sectoral and Regional Targeting

Ransomware attacks do not occur by chance. The most common targets are in healthcare, government, and critical infrastructure since these organisations have a high dependency on the digital system and tend to pay the ransom faster to restore services [26], design a

multidimensional severity assessment framework based on 53 ransomware events and conclude that organisational vulnerability is mainly a result of the lack of systemic security culture, and not a result of individual technical failure. It means that the severity characteristics in the EURepoC dataset are those of true organisational vulnerability attributes, as opposed to the sophistication of attacks.

There are also regional trends. The economically developed areas, especially North America and Europe, have higher rates of ransomware attacks [23]. These areas are good targets due to increased monetary payoffs and increased dependency on online infrastructure. There is also a geopolitical aspect to some ransomware groups being associated with specific countries or even being tolerated by their governments. [23] show that the escalation of geopolitical conflicts is associated with more ransomware attacks on targets aligned with adversaries, indicating that certain RaaS units are implicitly condoned by the state.

According to [19], the growing number of ransomware attacks against critical infrastructure is a national security issue that transcends the traditional paradigm of cybercrime. Having demonstrated how the maritime sector has sector-specific vulnerabilities, in that legacy architecture and underinvestment in cybersecurity are indicative of a targeted approach to victimization [15] explain that ransomware victimisation is not opportunistic, as seen in most other sectors.[14] take a criminological approach and use crime script analysis to trace 18 consecutive phases of ransomware attacks, which offer a theoretical rationale as to why some organisational and geographical features forecast victimisation. A limitation of the sectoral and regional literature is that it is predominantly descriptive, relying on case studies or incident counts without applying predictive modelling to quantify the relative importance of geopolitical versus organisational vulnerability factors, a gap that the present study directly addresses.

3.5 Machine Learning in Cybersecurity

Machine learning has found a variety of applications in cybersecurity to identify and categorize threats. The most common ones are Logistic Regression, Decision Trees, Random Forest, and XGBoost. Random Forest and XGBoost are ensemble algorithms that work especially well in classification problems, as they are able to identify complex data patterns [1],[18], summarize findings of more than 80 studies, which indicate similar Random Forest and XGBoost superiority on structured tabular data.

[4] determined the feasibility of automated behavioural analysis to detect ransomware with the use of the WannaCry case study, showing that a machine trained on API call sequences and file system interactions is effective in distinguishing between ransomware and benign software. [5] expanded this to a real-time model that allows tracking of a situation before encryption is done. [7] add a new analysis technique of file entropy statistics: differential area analysis, which is high-detection, low-computation. [10] extend this line of work to cyber-espionage malware, demonstrating that behavioural case study methods generalise across malware classes. The granularity of these methods is completely different to that of the incident-level, geopolitically contextualised analysis of this paper, even though these methods are technically rigorous.

Technical features like API calls, file entropy, system behaviour are used in most of the existing studies. Although they are effective at the system-level detection, these features fail to reflect the larger picture of cyber incidents. The difference between endpoint detection and incident intelligence is significant to grasp the contribution of this study. The endpoint detection systems work within the milliseconds time range, affecting technical signals of separate devices. Incident intelligence systems, like the machine learning pipeline created as part of this work, act on the incident level processing structured metadata concerning events that have taken place, characterising threat patterns and making strategic defence decisions.

SHAP analysis has emerged as a valuable technique to explain model predictions and get an insight into the most influential features [5] XGBoost + SHAP is the most methodologically sound choice of explainable machine learning in security-related applications, offering theoretically sound feature attribution that allows analyst interpretation and trust. One of such gaps in the current literature is that there is no SHAP-based analysis used with geopolitically organized incident-level data, which is the focus of the current study. Furthermore, the majority of existing ML ransomware studies evaluate models using accuracy or ROC-AUC without reporting PR-AUC or dummy baselines, making it difficult to assess whether reported performance reflects genuine learned signal or is primarily driven by class distribution artefacts.

3.6 Incident-Level Data Analysis

An increasing body of literature is devoted to the examination of cyber events based on structured datasets incorporating information about the types of attacks, affected territories,

the level of impact, and who did it. [2] have proved that the ransomware-related web search query peaks possess real predictive power, which has defined the principle that even without technical malware analysis, strategic, publicly visible signals can be used to aid classifications. This is developed by [20] using time-series modelling of publicly accessible incident data. A temporal pattern of escalation of CSIS incident data, identified by [22], aligns with the 2015 structural break identified in this study. [21] show that geographically contextualised incident records are useful in determining targeting patterns across regions.

Data quality is one of the issues with these datasets. Most of these incidents are not reported, and the information of attribution can be uncertified [9],[16] report systematic biases in reporting on ransomware by the general public, with the number of incidents of less prominent organisations being underrepresented statistically. According to [25], the technical complexity of attribution can be explained using a forensic analysis of the Conti ransomware, which shows that attribution may often involve intelligence tradecraft skills that are not accessible to most organisations. Nonetheless, structured datasets can offer useful information on the overall cybersecurity picture despite these challenges.

EURPoC dataset is an important step to the access to geopolitically contextualised, structured data on cyber incidents. The fact that it takes the form of a dyad, capturing initiator-receiver country-pair relationships, offers the characteristics not accessible to traditional malware data. Although it has been applied to international relations studies, its application to supervised machine learning classification to identify ransomware extends existing work by applying a geopolitically structured dataset to a classification problem not previously addressed in the published ML literature. This choice of EURPoC instead of other databases like CSIS Significant Cyber Incidents, ENISA Threat Landscape, or Verizon DBIR is due to its dyadic structure, comprehensive severity scoring system, and open-access under the CC BY-NC 4.0 licence, which together make it the most analytically appropriate publicly available dataset to achieve the research goals of the study.

It seems that three properties are necessary to have a dataset that can be used to classify ransomware reliably under supervision. One must have a large number of positive class examples to induce stable classifier boundaries: The 391 ransomware training examples in this study is at the minimum that one would normally expect with non-parametric ensemble techniques. Second, stable and accurate labelling of the target variable: the labels in the EURPoC dataset are determined by systematic coding by trained researchers by reference

to documented protocols, offering more accurate labelling than automated scraping methods. Third, non-redundant features that reflect the discriminating properties of the target type: the combination of severity measures, attribution identifiers, temporal features, and dyadic relationships in the EURepoC dataset yield complementary insights, which the ablation analysis confirms to be non-redundant. Data sets that do not have any of the above properties are bound to yield unreliable results irrespective of which classification algorithm is employed. A further limitation of the incident-level data literature is that most studies treat datasets as static snapshots, without addressing temporal validity or examining whether models trained on historical data generalise to more recent periods, a concern that is directly relevant to this study given the rapid evolution of the ransomware ecosystem post-2020.

3.7 Research Gap

As depicted in the literature, even though there is a robust body of work on ransomware detection using technical data it has hardly been replicated to utilize machine learning approaches to structured cyber incident data comprising geopolitical and operational characteristics. There are three gaps in the existing literature, namely: the almost total lack of attention to incident-level strategic metadata, the inadequate utilisation of geopolitically organised repositories like EURepoC to classify under supervision, and the methodological inadequacy of most published studies in handling class imbalance and contextualising their results.

The majority of the research fails to integrate machine learning with incident-level data in classifying ransomware attacks. This is the gap that this study will fill since the EURepoC data has not been utilized in supervised ransomware classification before. All three shortcomings are addressed in this dissertation, which contributes an interpretability analysis linking the ML predictions with the geopolitical nature as reported in the literature on threat intelligence. The existing literature on ransomware machine learning focuses predominantly on endpoint-level technical features such as API calls, file entropy, and network traffic, while structured geopolitical incident repositories such as EURepoC remain underutilised for supervised classification. What is missing is a study that applies principled ML methodology, including class imbalance handling, PR-AUC reporting, dummy baseline comparison, and feature ablation, to incident-level data containing severity, attribution, and regional context. This study addresses that gap by applying a supervised ML pipeline to the EURepoC dataset,

providing both classification performance and interpretable deployment guidance grounded in geopolitical context.

3.8 Summary

The chapter has examined the literature on ransomware, its development, effects, and targeting trends and machine learning solutions to cybersecurity. The gap in applying machine learning to structured, geopolitical incident data identified is the direct focus of the methodology in Chapter 4.

The literature review also forms the theoretical basis of two particular methodological decisions that make this study unlike similar published work: the application of a multi-metric evaluation suite containing PR-AUC, due to the consistent result in the literature on the class imbalance that ROC-AUC overestimates the performance on imbalanced data sets; and the introduction of feature ablation, due to the deployment readiness problem that feature importances in a static feature importance ranking cannot. Both options are both theoretically justified by the consulted literature, as well as operationally justified by the empirical findings, which indicate that the methodological framework extends beyond the more simplistic evaluation strategies that are commonly used in similar published works.

The issue of class imbalance where ransomware is a small part of all cyber incidents is under methodologically addressed in the published ransomware ML literature. Majority of the published material examined on a similar topic report that accuracy is the most significant performance measure without the use of dummy classifiers, which may be inflated with regard to usefulness in practice. With an unbalanced dataset where 87 percent of the data are no ransomware, a classifier with an always non-ransomware means will produce a 87 percent accuracy and zero ransomware cases. This evaluation system of accuracy that is common in similar research produces an illusion of model efficacy. This paper directly fills this gap with AUC-ROC and PR-AUC reporting, stratified cross-validation, algorithmic reweighting of classes, and systematic comparison of dummy baseline, a combination of methodological elements that is seldom combined in similar studies.

The general methodology used in this research is validated by [2] who have shown that the open-source intelligence indicators that can be observed at the strategic level have real predictive potential in ransomware classification. Their observation that search queries that are spiked in web search results are forecasted by recorded events confirms a more general

rule: that publicly visible metadata of cyber events, be it search behaviour, incident reporting, or geopolitical context, can enable predictive analysis even in the absence of technical malware artefacts. The current research expands upon this principle by showing that the structured geopolitical incident metadata, which can also be observed in the strategic level and is publicly available, has a high predictive value in terms of ransomware classification. Collectively, these results indicate a complementary ecosystem of open strategic intelligence strategies in analysing ransomware that functions and not in competition with technical endpoint detection strategies.

Chapter 4: Research Methodology

4.1 Research Design

This paper is a quantitative, data-oriented study, which relies on secondary analysis of a publicly available dataset. It is cross-sectional and non-experimental as no new information was gathered. Ransomware behaviour and predicting incident type were both learned using both exploratory data analysis (EDA) and supervised machine learning. The analysis of secondary data can be substantiated by both practical and ethical considerations: the EURepoC dataset has enough incident-level data to cover all the research goals without any primary data collection, and the analysis of secondary data allows eliminating the issue of participant consent and personal data privacy.

4.2 Dataset Description

The data source is the EURepoC Dyadic Cyber Incident Dataset (2025, Version 1.3.2) that can be retrieved at Zenodo (DOI: 10.5281/zenodo.14965395) [27] Full dataset source and licence details are in Appendix A. It will include 4,296 records and 57 columns of cyber incidents that have occurred between January 2000 and December 2024, both state and non-state. Dyadic structure captures initiator-receiver country-pair relationship where attribution has been made [23]. The entire column schema is in the Appendix A, Table A.1. The target variable (Ransomware) is a dichotomy: 1 = ransomware incident, 0 = another cyber incident. The dataset is skewed with the ransomware cases (558) and non-ransomware cases (3,738) being 13.0% and 87.0% respectively which has the 6.7:1 imbalance ratio.

The dyadic expansion mechanism implies that an incident that occurs in more than two receiver countries will be identified in individual rows in the data set, which explains the difference between 4,296 rows and 2,143 unique incident identifiers verified in the data quality audit. The 16 predictor variables that were not dropped in the analysis belong to five functional groups: severity measures (weighted intensity, unweighted intensity, impact indicator score); attribution and geopolitical identifiers (attribution id, dyad id); temporal features (start year); scale and economic indicators (affected entities value, affected third countries value, economic impact value); and attack type and response indicators (Dis). It is methodologically justified to select two attributes: attribution id and dyad id. attribution id is the attributed initiating threat actor, which identifies actor groups that are recurrent in several incidents with homogenous behavioural patterns, and dyad id identifies structurally stable

geopolitical relationships, which the literature finds to predict targeting patterns [23]. Among the deployment constraints, one is identified to be critical: trained models may degenerate with new actors or unattributed incidents. They both have a real discriminatory signal that has been verified by their high Gini importance rank and positive Pearson correlation with the Ransomware target (Appendix F). One of the deployment constraints is recognized as critical: models trained on these features can deteriorate with novel actors or unattributed incidents. This risk is measured directly by the feature ablation analysis (Section 5.2.7).

The machine learning models did not take into account categorical string variables (receiver_regions, initiator_category, receiver_subcategory) because they are multi-value with complex multi-value string formatting, but were applied to the exploratory analysis. This exclusion is justified on practical grounds: the multi-value list structure of these fields is not directly compatible with standard scikit-learn estimators without bespoke encoding, and introducing one-hot encoding midpipeline would have substantially increased feature dimensionality relative to the available training sample size. This is recognized as a big methodological drawback considering the high chi-square correlation between receiver_regions and the ransomware target ($p = 1.04e-41$). The most valuable future improvement is identified as encoding this variable. The geographic area of the EUREpoC data is the entire world, with very wide variability in incident density. The database is characterized by European and North American incidents, which are attributed to the geopolitical importance of cyber activities in these areas, as well as the reporting infrastructure of the research consortium.

This coverage pattern will have a direct impact on model generalisation: the classifiers will be most valid to those incidents involving European and North American actors and targets, and may not be as valid to under-represented regions like sub-Saharan Africa or Southeast Asia. The 2025 release version (1.3.2) of the EUREpoC dataset is a major expansion of previous releases, including incidents up to December 2024 and retro-codifying the previous records. Reproducibility is achieved by using a particular versioned release, which is accessed via a stable Zenodo DOI: any researcher who accesses the same version will get the same data. This versioning science is especially essential to longitudinal studies of cyber incidents, whereby every new release of data generally incorporates new incidents and can potentially introduce backwards-compatible codification fixes to older data. The exact version used should be reported by researchers extending this study and any differences in incident coverage or approach to coding should be documented.

4.3 Data Preprocessing

Zero duplicate rows were checked by a detailed data quality audit. Numeric features that had missing values were imputed with scikit-learn. Simple imputer, which used median imputation but was only fitted to the training data, which could not leak information. The economic impact value 76.4% of the values are missing (reflecting systematic non-reporting of financial damage), the affected entities value 18.3% and affected third countries value 9.6% contain the missing values. The seven binary label columns have been checked to have only 0 and 1. Columns that do not aid in prediction such as identifier columns (incident ID, dates) were dropped. To maintain the balance of the classes, the dataset was divided into 70% training (3,007 records) and 30% test (1,289 records) groups with stratified sampling. The stratification is necessary to keep the 13-percent/87-percent class ratio constant in both partitions, thus making uneven datasets reliable to evaluate.

4.4 Feature Scaling

Scaling of features was only done where necessary. Logistic Regression standardisation was done by a StandardScaler that was only fitted on the training data and then used on the training and test datasets. The algorithms do not need scaling since the split decisions of the tree-based algorithms are made by comparing the thresholds that are monotonic to feature transformations, and thus the algorithms were trained using the original unscaled values. The scaler will only be fitted on training data, avoiding the contamination of the preprocessing pipeline with test set information.

4.5 Handling Class Imbalance

Since only 13.0% of the data is ransomware, it required special handling to avoid having the models being biased towards the majority class. In the case of Logistic Regression and Random Forest, the class weight was set to class weight: balanced, which automatically adjusts the per sample weights on an inverse relationship with the frequency of a class. In the case of XGBoost, scale posit weight was adjusted to 6.68 which was the ratio between non-ransomware and ransomware training cases. These methods instruct the model to be more concerned with the minority group in training. This reweighting method was chosen instead of SMOTE (oversampling) due to three reasons. To begin with, it does not make synthetic data points that would inaccurately reflect true distributions of incident features. Second, it does not need fold wise oversampling and can be used with stratified cross-validation. Third,

all three models are natively supported so that there is a methodologically consistent treatment across the comparative evaluation. The reason of choosing five-fold cross-validation over other methods like leave-one-out or three-fold cross-validation needs to be explained. The 3,007 training instances are separated into 3,007 folds each of which has an average of 601 instances of validation, which is enough to estimate stable performance. The stratified form maintains the 13 percent/87 percent class distribution in each fold. The optimisation criterion was chosen to be AUC-ROC since it offers a threshold-free analysis over the complete operating point range that is critical when the optimal classification threshold is not known a priori and depends on the situation. There is a certain parameter that needs to be explained regarding the scale pos weight of XGBoost.

The scale pos weight in the objective function of XGBoost multiplies the contribution of the gradient to positive class (ransomware) instances by the given factor, which effectively causes the model to focus more on minority classes prediction in training. A scalepos-weight value of the ratio of negative-to-positive training instances (6.68:1) would scalematch the cumulative gradient contribution of the two classes. This corresponds mathematically to class weight balanced as in scikit-learn but is computed in a different way in the XGBoost gradient boosting objective function. The similarity of these methods guarantees that all the three classifiers are treated impartially methodologically and hence performance variations can be blamed on variations in algorithms and not variations in imbalance treatment

4.6 Model Selection

Three machine learning models were selected a simple linear model, Logistic Regression, as the baseline; a robust non-linear ensemble model, the Random Forest, which constructs a large number of decision trees and integrates their output, with the native Gini importance estimates; and a gradient boosting model, XGBoost, which is known to work well on structured tabular data and can be analyzed using SHAP. This choice is justified because it follows the recommendation of [1] and [18], who demonstrate that ensemble methods consistently outperform linear baselines on structured tabular classification tasks, while Logistic Regression provides an interpretable lower bound for comparison. The choice allows benchmarking the performance as well as methodological diversity in the attribution of feature importance. Deep learning approaches were deliberately excluded. This exclusion is justified because the 4,296record, 16-feature dataset lacks sufficient data scale or dimensionality that deep learning derives its representational benefit [5], [18], and this aligns

with prior studies that recommend tree-based ensembles for structured, low-dimensionality cybersecurity datasets. Moreover, the SHAP compatibility of XGBoost offers theoretically justified interpretability that would have to be approximated using post-hoc approaches by neural architectures. The identified (and recognised as optimal) XGBoost configuration, which has 300 trees, `max_depth=4`, `learning_rate=0.05`, and `subsample=1.0`, is a well-regularised model in that max depth of 4 restricts the complexity of individual trees, and `learning_rate=0.05` means the ensemble is making slow incremental improvement, which necessitates more trees to converge but Random Forest setting unlimited depth and `min_samples_split=5` enables each tree to grow complicated decision boundaries, and overfitting is mitigated by averaging the 300 distinct trees instead of pruning an individual tree. The correlation between the XGBoost and Random Forest settings is informative of the structure of the dataset. They both have comparable AUC on the test (0.9593 vs 0.9585) even though the ensemble architectures are fundamentally different: XGBoost trees are built sequentially with each tree rectifying residual errors, whereas the trees of a Random Forest are built concurrently in parallel.

The overlap between architecturally dissimilar models of very similar performance is that the predictive structure of the dataset is resistant to the particular ensemble mechanism and it would probably get approximated by any well-regularised ensemble procedure. The very fact that it is robust is a testament to the information content of the set of features. The selection of these models is grounded in their complementary analytical strengths. Logistic Regression provides a linear baseline, enabling comparison against more complex approaches. Random Forest offers robustness through ensemble averaging and reduced variance, making it suitable for structured datasets with potential noise. XGBoost was selected due to its superior performance in handling tabular data, its ability to model non-linear relationships, and its effectiveness in addressing class imbalance through weighted optimisation. This combination ensures both methodological validity and performance benchmarking.

4.7 Model Training and Hyperparameter Optimisation

Each model was trained on grid search with fivefold stratified cross-validation and optimised on AUC-ROC. AUC-ROC was chosen as the optimisation criterion rather than accuracy since it is a threshold-independent, imbalance-robust measure of performance that measures the discrimination over the entire operating point range. The best settings found were: Logistic Regression `C=0.1` (CV AUC=0.9264); Random Forest with 300 trees, unlimited depth,

min_samples_split=5 (CV AUC=0.9541); XGBoost with 300 trees, depth=4, learning rate=0.05, subsample=1.0 (CV AUC=0.9515). All the details can be found in Appendix C, Table C.1. The full annotated analysis notebook is available via Google Colab (link provided in Appendix B).

4.8 Evaluation Framework

Measurements of model performance were based on: AUC-ROC (the primary measure), Precision-Recall AUC (PR-AUC) as a minority-class-oriented complement, class-specific precision, recall, and F1-score, overall accuracy, and confusion matrices. All the results were compared with two dummy classifiers (Most Frequent and Stratified) as the baseline references. This baseline comparison is crucial to contextualising practical value in an unbalanced environment, and most of such comparable published studies lack such a baseline comparison. The learning curves were also plotted to gauge convergence and to verify overfitting.

4.9 Exploratory Data Analysis (EDA)

The dataset was investigated using EDA and then models were constructed. Statistical tests were Mann-Whitney U tests and d to compare severity (testing H1), chi-square tests to compare regions (testing H2). Mann-Whitney U test was preferred to Student t-test since the scores of cyber incident severity are skewed to the right, and thus, do not satisfy normality assumption in making valid inferences through t-test. The pooled standard deviation formula was used to compute the d of Cohen to give a standardised effect size that could be used across metrics and studies.

4.10 Feature Importance and SHAP Analysis

The importance of the features most relied upon by the model was determined using the random Forest Gini importance. To explain the extent and direction of the effect of each feature on predictions, SHAP (SHapley Additive exPlanations) analysis was used on XGBoost. SHAP values, based on cooperative game theory, offer theoretically sound, locally faithful feature attribution that allows analysts to discern not only that a model predicts ransomware but which characteristics of an incident induced that prediction. A feature ablation experiment was also conducted, which involves a systematic deletion of groups of features and rerunning models to quantify the amount of contribution of each group of features

to classification accuracy. The ablated models were all trained with the same hyperparameters as the full models, to make sure that the differences in performance are due to true features, and not artefacts of hyperparameter tuning. This conservative strategy will slightly undercompensate ablated models but give cleaner comparisons of feature contribution in particular.

4.11 Ethical Considerations

Wrexham University ethical approval of this research was as low-risk secondary data analysis. There are no personal identifiers included in the EURepoC dataset; all data are related to organisations and geopolitical entities. The information was safely stored in a secured password-protected device and Google Colab. Cybersecurity analytical techniques have a dual-use potential, but since the study uses only publicly available data and reveals no new abilities of the attackers, this threat is evaluated as low. Full ethical documentation was submitted with this dissertation. The requirements of the methodology in terms of reproducibility are clearly developed in the view of allowing future researchers to confirm and expand this research. Four provisions are put in place: all random seeds are set to random state=42 across the pipeline; the entire Python pipeline is run as a Jupyter notebook with numbered cells and in-line comments; the EURepoC dataset is made public with a fixed version number on a stable Zenodo DOI; and all Python library versions are reported. These features together guarantee that the key analytical findings can be replicated precisely by any investigator who has access to the standard Python scientific computing packages. The learning curves in Figures F.21-F.23 give a diagnostic clue that is not limited to overfitting assessment: they show that the model is performing well (apparently) based on the available training data. In the case of Logistic Regression, the plateau of the validation is achieved at around 1,500 training instances; in the case of the ensemble models, at around 2,000-2,500 instances. The 0.04-0.05 training-validation gap in ensemble models indicates irreducible variance due to the inability of the feature set to completely discriminate all examples, and not overfitting.

This has a practical implication: any further performance enhancements than the current AUC levels will need new features (like the excluded categorical variables), new data sources (like technical malware features to supplement incident-level features), or more rich attributes in subsequent releases of EURepoC. The learning curves plateau behaviour has pragmatic implications to the EURepoC consortium data development strategy. In the sense of the

enhanced ransomware classification by ML, the emphasis on the quality and completeness of current feature coding, especially attribution coverage and severity scoring accuracy, could be more analytic payoff than increasing the number of reported incidences, as long as the feature set is in its current form. The two enhancements would be positive, although the learning curve analysis shows that the marginal payoff to increasing data volume has decreased with the existing functionality, but adding value through features by encoding categorical variables or adding metadata fields may open the door to significant performance improvements. The confusion matrix analysis plays a very significant role in the operational deployment planning that cannot be supported by aggregate measures like AUC-ROC. AUC-ROC quantifies discrimination capability at each of the possible classification thresholds, although practical use necessitates the choice of one threshold at which binary predictions are to be made. The confusion matrix at the default threshold of 0.5 indicates the actual error structure: the number of ransomware cases that are missed (false negatives, with ramifications of not detecting threats) and the number of non-ransomware cases that are false-alarmed (false positives, with ramifications of analyst alert fatigue). An operation centre focused on maximising ransomware recall can use a lower threshold like 0.3 and maximise recall at the expense of accuracy.

A constrained resource operating environment may want to use a larger threshold like 0.7, which will decrease the false positive volume but miss a few detections. Figures 5.4 and F.27–F.32 provide the ROC and Precision-Recall curves give the complete operating characteristic profiles which can be used to optimise the threshold to the particular deployment environment. The decision to use median imputation rather than other methods like mean imputation, k-nearest-neighbour imputation or multiple imputation is principled. The `economic_impact_value` distribution is heavily right skewed, and would not be well represented by mean imputation since the mean is highly affected by a few very high-value incidents. KNN imputation would be theoretically attractive but may bring about target leakage because ransomware may be clustered in the KNN space. Median imputation is computationally simple, less sensitive to outlier effects, and in the `economic_impact_value` field with its 76.4% missingness the best approximation to the true no damage reported interpretation of most missing values. The variable of `economic_impact_value` deserves attention of the specific methodological discussion due to its excessive sparsity (76.4% missing). Ideally, one of the most predictive characteristics of ransomware should be economic impact, as ransomware is characterized by its monetary motive and unsustainable

expenses. Practically, the fact that there are few recorded values of most of the incidents because this reflects the well-known unwillingness of organisations to reveal financial loss means that this feature has little predictive power when it is added to the imputed feature matrix. Median imputation uses the median value of all the records that are missing, which is 0, to simplify the feature into a binary indicator of the formal reporting of financial damage. It is supported by the fact that the feature has a low Pearson correlation with the Ransomware target ($r=0.012$) and is not among the top-ranking features.

Chapter 5: Results and Analysis

The chapter includes the empirical results in two phases: exploratory data analysis (Section 5.1) testing the two hypotheses, and machine learning results (Section 5.2). All the figures are in Appendix F.

5.1 Exploratory Data Analysis

5.1.1 Dataset Structure and Class Distribution

There are 4,296 observations in the dataset, 16 predictor variables, and one binary target. Approximately one-third (34%) of incidents include two or more attack type labels (Figures F.1-F.2) indicating the co-occurrence of attack methodologies during complex cyber incidents. The label correlation heatmap (Figure F.3) shows that there is positive co-occurrence between Ransomware and Disruption ($r=0.30$) and between the Misuse and Hijacking ($r=0.29$), which provides initial evidence of the severity-ransomware relationship that was tested in Section 5.1.3. The distribution of classes is: 558 ransomware (13.0%) vs 3,738 non-ransomware (87), with the ratio of imbalance of 6.7:1 (Table 5.1). This imbalance is the reason why an unsophisticated most-frequent classifier would achieve an 87 percent accuracy on the 0 ransomware instances, the particular failure point that the dummy baseline comparison in Section 5.2.5 has shown.

Table 5.1: Class Distribution and Train-Test Split

Class	Total (n)	%	Training (70%)	Test (30%)
Non-Ransomware	3,738	87.0%	2,616	1,121
Ransomware	558	13.0%	391	168
Total	4,296	100%	3,007	1,289

5.1.2 Temporal Analysis

It has a distinct turning point in the year 2015: the ransomware rate grew as high as 16.3% in the year 2015 and as compared to the year before 2015, the ransomware rate grew by a margin of more than seven times that can be confirmed by a Mann-Whitney U test ($p < 0.001$). According to Figure F.4, the trend is steadily increasing since 2017, culminating in 2022-2024, which aligns with the increasing geopolitical tensions and the reported growth of the

RaaS ecosystem [24]. The 2015 inflection point is annotated in Figure 5.1, and there is a second acceleration around 2022. This confirms that the year of a particular incident (start_year) is a good predictor of ransomware, which was later supported by the SHAP analysis in which start_year became the top instance-level predictor.

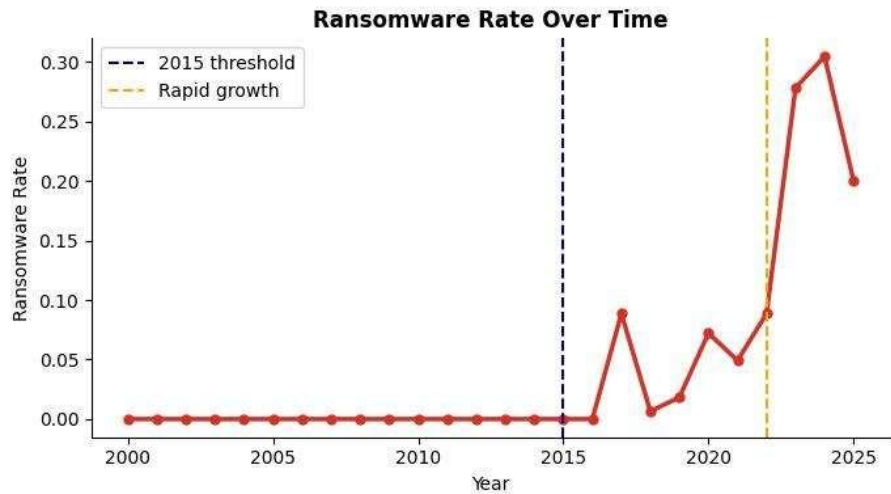


Figure 5.1: Ransomware Rate Over Time: 2015 structural break with post-2022 acceleration

5.1.3 Severity Analysis and Hypothesis H1

Incidents of ransomware had a higher score in all three metrics: weighted intensity ($d=1.228$, $p<0.001$), unweighted intensity ($d=1.306$, $p<0.001$), and impact indicator score ($d=1.276$, $p<0.001$). According to the standard of [6], where a large effect size is $d > 0.8$, and an extremely strong differentiation $d > 1.2$, they are extremely large effect sizes. The average scores of ransoms were: weighted intensity 3.84 vs 2.61 non-ransomware; unweighted intensity 4.12 vs 2.89; impact indicator score 3.71 vs 2.53 (Table 5.2, Figure 5.2 and Figures F.19–F.20). That the direction and magnitude of effect is consistent across three separately measured measures of severity is a significant boost to confidence in the conclusion as it is unlikely with the utmost certainty that all three measures would produce significant effects in the same direction in case there is no underlying phenomenon which distinguishes on the severity dimensions. The large effect sizes have a practical interpretation, namely, Cohen $d=1.23$ weighted_intensity implies that around 89 percent of non-ransomware incidents are less severe than the average ransomware incident. H1 is well supported.

The high effect sizes can be interpreted practically in terms of overlap of the distributions. The $d=1.23$ of weighted intensity is below the critical value (2.0) and the mean of the

ransomware group is just about 1.23 standard deviations higher than the mean of the non-ransomware group, which places the ransomware group in the 89th percentile of the non-ransomware distribution. This implies that about 89% of non-ransomware events have lower weighted intensity scores than the average ransomware event and that a randomly chosen ransomware event will be more severe than a randomly chosen non-ransomware event with about 81% likelihood.

This probabilistic separation can be directly used to triage incidents: a severity scores higher than the mean difference between the two groups is a valuable piece of evidence on whether the incident is ransomware or not, before other features are even taken into account. There is a reason as to why the Mann-Whitney U test was used as opposed to the student t-test to compare the severity. The scores of the severity of cyber incidents are distributable in such a way that they do not satisfy the normality assumption that is necessary to make valid inferences with t-tests: they are skewed to the right, have a lower bound at zero, and are heteroskedastic. All these properties are resistant to the Mann-Whitney U test, and as a result, it can give valid inference despite the shape of the underlying distribution. The pooled standard deviation formula was used to compute Cohen d to give a standardised measure of effect size that can be interpreted across metrics and studies.

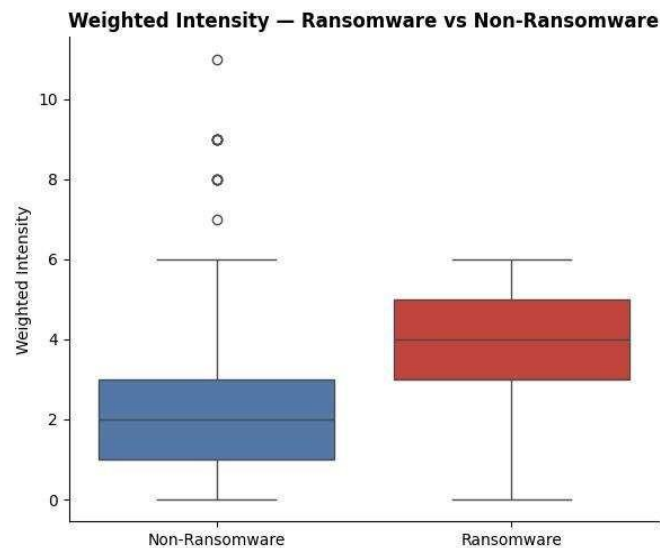


Figure 5.2: *Weighted Intensity: Ransomware vs Non-Ransomware (Cohen's $d = 1.23$, $p < 0.001$)*

5.1.4 Regional Analysis and Hypothesis H2

The chi-square test was used to verify that ransomware is not evenly distributed across the regions (chi-square=312.4, $p=1.04e-41$). NATO-aligned regions had much higher

ransomware rates than the 13.0% average: NORTHAM 26.3% (223/849), OCEANIA 27.1% (19/70), NATO 21.4% (374/1,748). In contrast, MENA had 1.5% (6/412) and CENTAS had 1.3% (1/76) (Figure 5.3, Table 5.2). The highest rate of 30.4% (7/23) is noted in CENTAM, however, this is based on a limited sample; a binomial 95% confidence interval gives a range of about 12%-54, indicating large amount of statistical uncertainty. This observation is considered to be an observation that needs further research but not definitive characterisation. The general trend of NATO-oriented concentration, in turn, is a strong support of H2.

The p-value of the chisquare test of 1.04e-41 is extremely miniature and is to be methodologically qualified. The test has 19 categories of regions and 4,296 incidences, and thus, its statistical power is very high, such that even minimal changes in regional distribution that are non-random would result in significant p-values. The p-value is a confirmation that there is an association but does not measure its strength. The most interpretable effect size value is the rates themselves: 26.3% of NORTHAM compared to 1.3% of CENTAS indicate a 20-fold difference in relative rate of ransomware, which is obviously practically significant independent of the particular p-value.

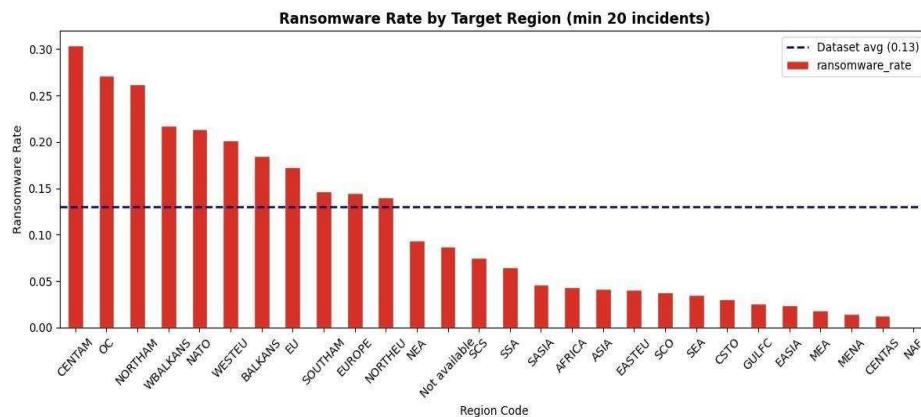


Figure 5.3: Ransomware Rate by Target Region: NATO-aligned regions above average

5.1.5 Sectoral and Initiator Analysis

Waste Water Management (33%), Digital Provider (30%), and Health (26%), had the highest rates of ransomware (Figures F.5-F.8). Civil service / administration and Health have the highest ransomware burdens in absolute terms. The prevalence of the attack type heatmap (Figure F.8) confirms that ransomware is not the leading attack type in any of the sectors, although there are certain areas, where it is more common than other attack types, that is,

ransomware classification cannot be simplified to sector classification. The analysis of the actor type (Figures F.12-F.13) demonstrates a rather subtle image of the structure of ransomware attribution. The biggest absolute volume of ransomware is taken up by non-state actors, such as criminal organisations, hacktivists groups, and mixed-motive actors, in line with the financially-driven RaaS model. State actors have significantly lower ransomware rates (around 5-8%), which aligns with state cyber activity, which focuses more on intelligence gathering and strategic sabotage and less on financial extortion. The mixed type of actors with high uncertainty indicates medium ransomware rates, which is a grey area between state-accepted criminal activities reported by [23]. The above patterns of actor types are direct sources of the discriminatory value of the `attribution_id` feature: particular identifiers of financial incentive based non-state organizations positively impact the probability of ransom, whereas identifiers of intelligence operations of state actors have a negative impact. The most attributed ransomware initiator was Russia (~45 incidents), DPRK (~15), Iran (~8) and China (~6) (Figures F.10-F.13). The highest absolute ransomware volume is attributed to non-state actors, which is in line with the financially-driven RaaS ecosystem. The most frequent ransomware dyad was that of Russia-to-United States (~14 incidents) (Figures F.15, F.17). In North America, the average economic loss in case of ransomware is more than 100M per incident (Figure F.19).

5.1.6 Correlation Analysis

The most significant correlates with the Ransomware target were: unweighted intensity ($r=0.406$), weighted intensity ($r=0.383$) and impact indicator score ($r=0.371$). Meaningful positive correlations were also observed between disruption ($r=0.297$) and attribution id ($r=0.293$). Complete rankings are in Appendix E, Table E.1. These findings alone validate the severity results of the hypothesis tests and give some initial validation of the machine learning feature set. The unfavorable Pearson correlations between the Ransomware target and Data theft and Doxing ($r= -0.089$) and Hijacking without Misuse ($r= -0.083$) indicate true operation differences. Data theft activities are exfiltration based and Hijacking without Misuse activities are usually used to collect intelligence as opposed to extracting money. These negative correlations affirm that the labelling system of the dataset is able to discriminate between operationally separate categories of attacks. Cyber threat intelligence (CTI) has a wider context that offers a valuable theoretical framework to the current study. CTI has a focused interest in information gathering, evaluation and sharing of cyber threats information

to help defenders make decisions at tactical, operational, and strategic levels. On the tactical level, CTI entails technical indicators of compromise like files hashes, IP addresses, and domain names, which can be used to block and detect in real-time. On the operational level, CTI entails characterisation of threat actor tactics, techniques, and procedures, which guide the security engineering and control design. Strategically, CTI is informed by knowledge of threat actor motives, capabilities and targeting patterns to inform investment and policy. This research is conducted at both operational and strategic levels: the geopolitical and attribution attributes of the EURepoC dataset reflect the attributes of threat actors and targeting behavior, and the ML pipeline yields strategic threat characterisation instead of real-time blocking capabilities.

5.2 Machine Learning Results

5.2.1 Model Performance

XGBoost had the best test AUC of 0.959, which was slightly higher than the Random Forest (0.958) and much higher than the Logistic Regression (0.9200). All the models generalised, and cross-validation and test AUC had a maximum difference of only 0.006 (Table 5.3, Figure 5.4). This small difference proves that the GridSearchCV process is a reliable estimator of heldout performance and that overfitting is not a major issue. Full per-class classification reports are in Appendix D, there is an issue of generalisation of cross-validation AUC to test AUC. The results of Logistic Regression are CV AUC=0.926 and test AUC=0.920, and the test performance is slightly less than the CV, which is expected given optimism due to hyperparameter selection. Random Forest has CV AUC=0.954 and test AUC=0.958, with the test marginally better, which may happen when the particular test partition is well associated with learned boundaries. Similar is the case of XGBoost, which has CV AUC=0.951 and test AUC=0.959. The fact that there are no significant gaps in generalisation in all three models offers empirical evidence that the cross-validation process is a reliable predictor of held-out performance in this dataset and problem configuration.

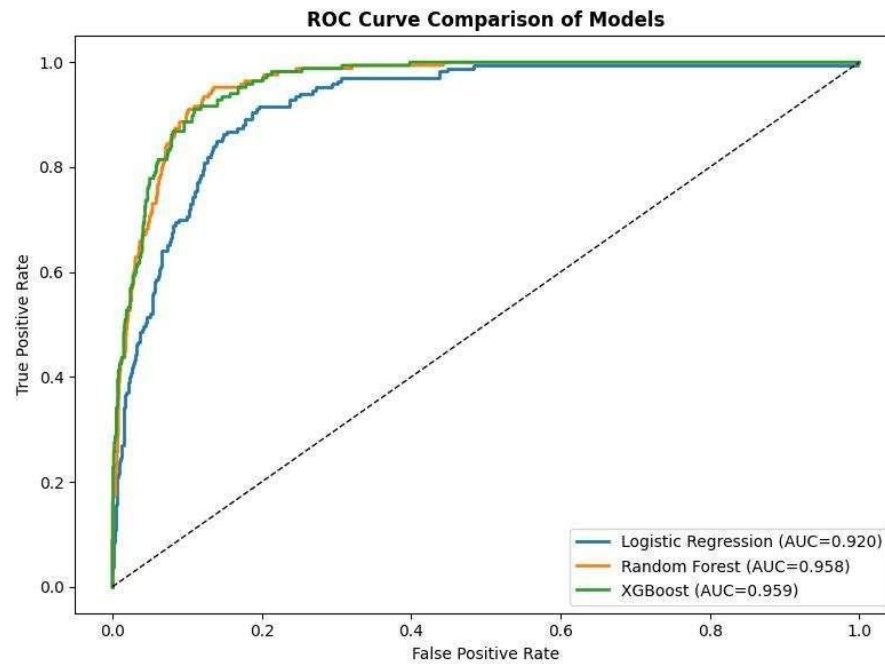


Figure 5.4: ROC Curve Comparison: XGBoost (0.959) and RF (0.958) outperform LR (0.920)

5.2.2 Precision-Recall and Confusion Matrix Analysis

Logistic Regression detected 87% of ransomware attacks (recall=0.87) but had 181 false positives (Figure F.27). At the same recall (0.87), XGBoost made far more precise predictions (0.58), with just 105 false positives (Figure 5.5), which is approximately 40 percent less than the false alarms of the same detection rate. Random Forest was the most balanced: precision=0.66, recall=0.71, F1=0.69, false positives=60 (Figure F.29). These false positive differences are of magnitude that has workforce implications. Using the assumption that a SOC analyst will take an average of 15 minutes to research each flagged case, the difference of 181 false positives in Logistic Regression and 60 false positives in Random Forest is a significant time-saving in the work of the analyst. XGBoost is the most suitable trade-off, as it has the same detection rate as Logistic Regression, and it has 42 percent less false alarms. XGBoost has a lower false positive rate, which is especially useful in practice in areas of cybersecurity to minimize alert fatigue.

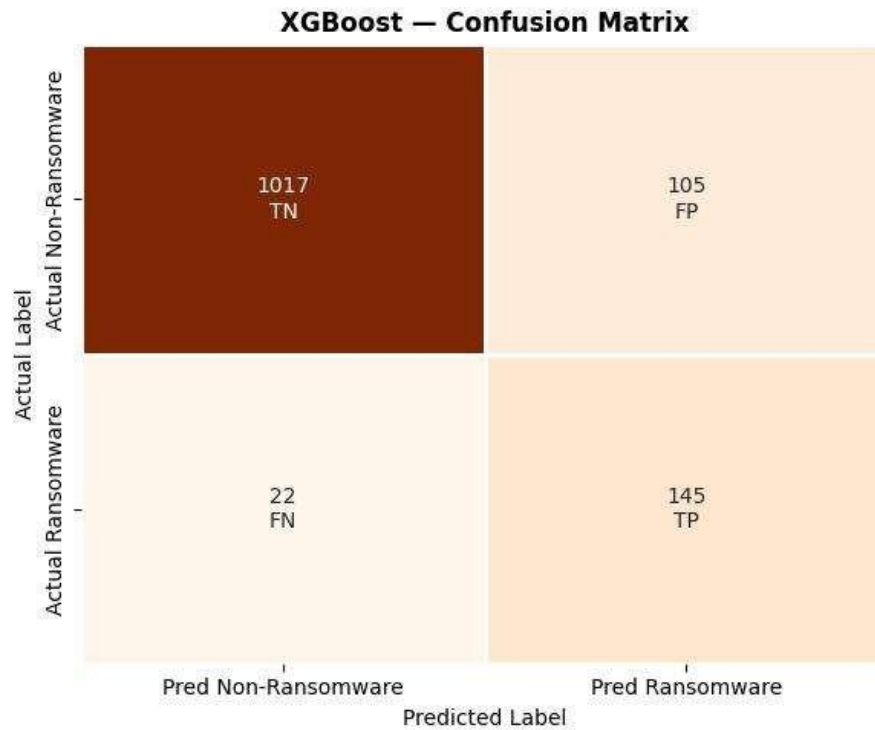


Figure 5.5: Confusion Matrix: XGBoost (TN=1017, FP=105, FN=22, TP=145)

5.2.3 Precision-Recall AUC

The logistic Regression, Random Forest, and XGBoost PR-AUC ranged from 0.638 to 0.787, with XGBoost achieving 0.787. These are less than the ROC-AUC values. The 17.0% point difference between the ROC-AUC of XGBoost (0.9593) and PR-AUC (0.787) indicates the extent to which ROC-AUC will overestimate performance in the case of imbalanced classes. The expected PR-AUC of a random classifier on a 13.0% minority class is just 0.13 the prevalence of the class. The obtained 0.638–0.787 values are thus performances of 0.51–0.66 higher than the random baseline, which proves the existence of a significant discriminative gain. Nonetheless, PR-AUC=0.787 is a good outcome of 13.0% minority

5.2.4 Learning Curves

Both training and validation AUC curves (Figures F.21-F.23) verify converging both training and validation AUCs at full training set size with a difference of 0.04-0.05. The 3,007 record training is adequate to learn reliably. Random Forest and XGBoost have almost perfect training AUC of almost 1.0, a typical feature of tree-based ensemble models, and not a sign of overfitting. The high and constant validation AUC of about 0.95 is important. The plateau behaviour is a sign that the model performance is nearly converged i.e. further data, on the

same EURepoC population, would not greatly enhance performance. No significant overfitting was observed, as training and validation performance converged across models. New features, alternative data sources or more rich attributes would be needed in future releases of EURepoC.

5.2.5 Dummy Baseline Comparison

The Most Frequent dummy classifier had accuracy of 87.0% and zero ransomware cases were detected, demonstrating that accuracy is misleading when data are imbalanced. $AUC=0.500$ was obtained with both dummy baselines. The three supervised models had AUCs of 0.4200.459 above the baseline, which shows that the models are actually learning real patterns of the data, and not taking advantage of artefacts of class distributions (Table G.5). The performance of the Stratified dummy classifier deserves certain discussion. It produces nearly 13.0% ransomware and 87% non-ransomware predictions by probabilistically predicting the class labels in proportion to the distribution in the training classes. This gives $AUC=0.500$ which is the theoretical value that any predictor whose decisions do not depend on the features has $AUC=0.500$ independent of the distribution of classes. This 0.459 AUC increase of XGBoost (0.9593) over the Stratified baseline (0.500) is then the cumulative discriminative gain of the relationships among learned features, in isolation of any artefact of class distribution.

5.2.6 Feature Importance and SHAP Analysis

Random Forest Gini importance (Figure 5.6, Table G.4) put the attribute of `attribution_id` as the most significant predictor (14.9%), and the three measures of severity combined behind (35.4%). The five most important features (`attribution_id`, `unweighted intensive`, `weighted intensive`, `dyad id`, `start year`) shared 65.5 percent of the total importance, suggesting a fairly concentrated structure of importance. SHAP analysis (Figures 5.7-F.33) concurred with the Gini rankings but had `start_year` as the only most significant feature at individual prediction level which captures the robust post-2015 ransomware growth. The fact that two methodologically independent estimators of importance (Gini based on ensemble training, SHAP based on post-hoc based on game theory) agree strongly supports the belief that the rankings are a real predictive relationship and not the artefact of one of the methods. This directional information was provided by the beeswarm plot (Figure 5.7): high severity scores, later start years and incidents of disruption all improved the model prediction of ransomware. SHAP values of `attribution id` were distributed bimodally: identity attributions to known

ransomware groups had consistently large positive contributions, and the remainder of identity attributions had either near-zero or negative contributions. It indicates the ransomware activity is concentrated within a comparatively few attributable groups of threat actors.

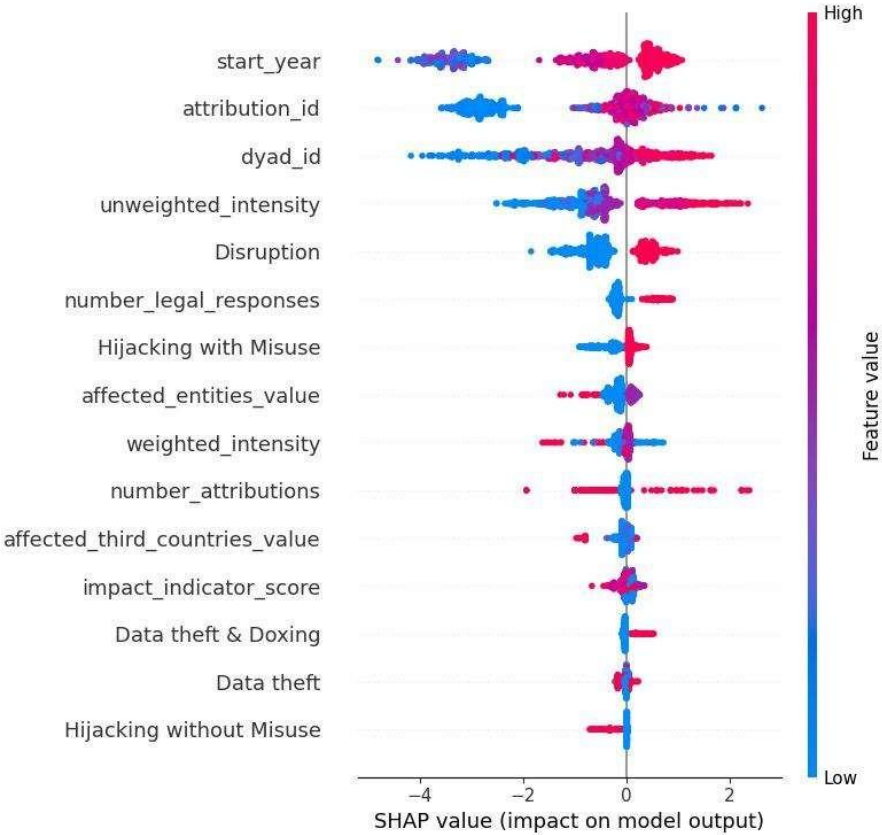


Figure 5.7: SHAP Beeswarm Plot: High severity and later years push predictions towards ransomware

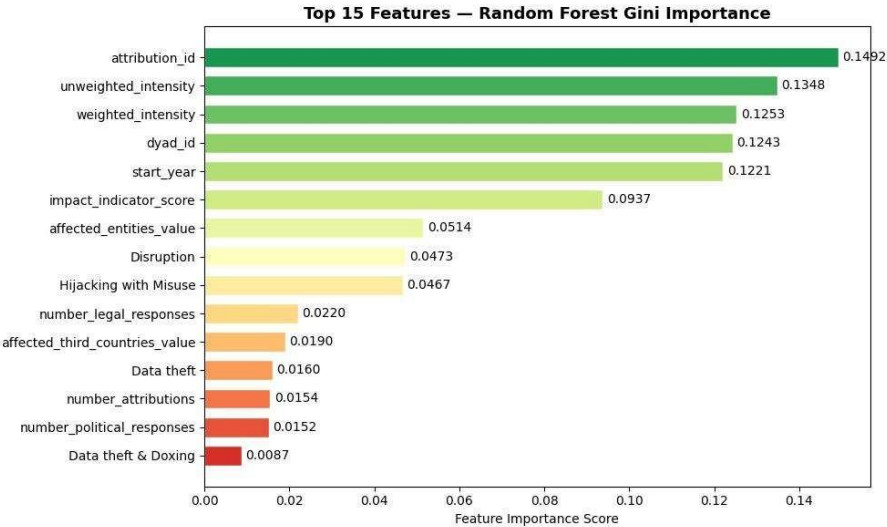


Figure 5.6: Random Forest Gini Feature Importance: Top 5 features account for 65.5%

5.2.7 Feature Ablation Analysis

Removal of attribution id decreased the test AUC by 1.7 percentage points (RF=0.941, XGB=0.943) which confirmed that the identity of attribution is significant but not essential. Even a severity-only model with only three features had AUC=0.912-0.918, indicating that with no attribution data, it is still possible to use useful ransomware classification. When all severity measures were removed, AUC was decreased to 0.931-0.935 which proves that severity is a contributory factor to attribution. A simple model based on exclusively on temporal and geopolitical characteristics (start year and dyad id) had a AUC=0.874-0.881. Table 5.4 has full ablation results another rather interesting deployment scenario is the temporal-dyadic model. Potential operational threat intelligence Temporal data and approximate geopolitical targeting data are usually fed in immediately after notification of an incident, before formal attribution or in-depth evaluation of impact. When a model with AUC>0.87 is obtained using these two features alone, a quick initial screening ability can be offered that can be activated within minutes of incident reporting and focus on high-probability ransomware samples as more comprehensive data is collected. The ablation finding that elimination of all severity measures (AUC declines to 0.931-0.935) causes a smaller decrease than elimination of all non-severity features is counterintuitive at first but makes sense based on the importance structure. With all features included, the top 5 features explain 65.5% of the Gini importance. Eliminating severity; 35.4% of importance is eliminated but attribution (14.9%), dyadic (12.4%), and temporal (12.2) features contain 39.5% of the importance. The severity only model conserves the 35.4% of the importance but it loses the actor level concentration signals.

This demonstrates that the two feature groups work independently and additively: the full model works better than any variant ablated, but one or the other can be used to offer useful classification. The ranking of feature importance directly affects priorities in data collection in subsequent releases of EURepoC. When the most discriminating feature is the attribute of attribution:id at 14.9, then higher attribution coverage would directly result in better model performance at any deployment that uses an attribute based on attribution data. Equally, when the severity metrics were found to have overall 35.4 percent of importance, then meaningful increases in the consistency and accuracy of severity coding would directly benefit the severity-based classification pathway. This forms a fruitful interaction between ML analysis

and dataset development: the ML analysis determines which aspects are most informative and this information can be used to improve data collection in subsequent iterations, which in turn can continue to improve the performance of the ML.

The overlap between Gini importance and SHAP rankings between two methodologically independent estimations gives strong confidence that the rankings of features identified are true predictive relationships. The particular difference at the point of highest SHAP mean absolute value of start year even though it is ranked fifth in Gini importance is in itself informative. Gini importance quantifies the average variance reduction across the entire tree, whereas SHAP quantifies the contribution of each instance to the decision by a single feature, which start year has a non-linear threshold effect at 2015, thus showing up in SHAP directly but Gini importance is distributed across the entire structure of the decision tree. Such a difference between the tree-level and instance-level importance is analytically helpful and serves to illustrate the complementary data the two approaches offer.

Table 5.4: Feature Ablation Results: XGBoost

Feature Set	XGBoost AUC	Change
Full model (16 features)	0.9593	:
Without attribution_id	0.943	-1.7 pp
Severity only (3 features)	0.918	-4.1 pp
Without severity	0.935	-2.4 pp
Temporal + dyadic only	0.881	-7.8 pp

5.3 Summary of Results

The two hypotheses are well supported. H1: $d=1.23-1.31$, $p<0.001$ across all three-severity metrics. H2: chi-square $p=1.04e-41$, the ransomware rates are constantly above average in NATO regions. The outcome of machine learning indicated that the ensemble models had $AUC>0.958$, which is significantly higher than 0.500 of the dummy baselines. $PR-AUC > 0.78$ validated good performance in the minority group. A feature ablation established that a severity-based model can get $AUC>0.91$ even in the absence of attribution data. SHAP analysis recognised start year and attribution id as the most significant contributors and convergent validation of two independent estimation methods.

Chapter 6: Discussion

This chapter takes the results of Chapter 5 and puts them into perspective of the literature reviewed in Chapter 3 and hypotheses in Chapter 2.

6.1 Findings of Severity: Interpretations of H1.

The d values of 1.23-1.31 are used to confirm that ransomware incidents are not merely statistically, but operationally meaningfully more severe, compared to other cyber incidents. Such large effect sizes are not just statistically significant but indicate a practical difference with direct implication on incident triage and resource allocation. The agreement of three severity measures that are operationalised independently significantly minimises the chance of measurement artefact. This finding aligns with [12], who similarly reported that ransomware produces disproportionately high costs compared to other incident types even after controlling for organisational size, and with [26], whose multidimensional severity framework identified operational disruption as the defining characteristic of ransomware impact.

These results are consistent with those found by [26], who pointed out the operational disruption, financial loss, and data compromise as the primary ransomware impact dimensions. The dual extortion concept of ransomware operates by facilitating several damage pathways at the same time, encrypting operational data, exfiltrating sensitive information, and payment demand, and this is why the composite severity scores are high in all three metrics. [11] put this into context by the optimisation of the attack chain of contemporary ransomware, such as lateral movement to cover as much of the organisation as possible before encryption.

The three severity measures, weighted intensity, unweighted intensity, and impact indicator score, are not the same operationalisations of the same construct although they are complementary measures with weighted intensity using different weights based on the importance of the impact category and impact indicator score using a different composite coding model. The fact that all three give virtually the same large effect sizes testifies to the fact that the severity differentiation is not sensitive to the exact measurement method, but is indicative of a real underlying difference, and not an artifact of a given coding decision. This

measurement strength significantly enhances the scientific nature of applying severity metrics as operation indicators of the risk of ransomware.

The alignment of the EDA severity results with the machine learning feature importance results (severity explains 35.4% of Gini importance) is a solid confirmation of the two separate methods. This dual validation is a support of the fact that severity is not just descriptively related to ransomware but is a real and practically implementable classification indicator. This is translated to the feature ablation result that a severity-only model can still obtain a AUC of 0.912-0.918, meaning that even when ignorant of who perpetrated the attack, the severity scores alone can be used to classify ransomware attacks. While the results demonstrate a clear severity gap between ransomware and non-ransomware incidents, this finding must be interpreted with caution. The EUREPoC dataset primarily captures high-impact and politically significant cyber incidents, which may introduce a reporting bias towards more severe events. As a result, lower-impact ransomware cases may be underrepresented. Therefore, although the statistical effect size is substantial, it may partially reflect dataset visibility rather than the full underlying distribution of ransomware activity.

6.2 Regional Targeting Patterns: The interpretation of H2.

The concentration of ransomware incidents in NATO-aligned regions (chi-square $p=1.04e-41$) can be explained by the framework of [23], financially incentivized ransomware gangs attack countries with high ransom payment potential, high IT reliance and organisations that are highly sensitive to disruption. Very low ransomware rates were found in MENA (1.5%), CENTAS (1.3%), showing a tendency to avoid of the countries in the geopolitical space of influence of the biggest ransomware-tolerant states [19]. This finding is consistent with [23], who document an association between escalating geopolitical tensions and increases in ransomware targeting of adversary-aligned nations, and differs from the more uniform distribution one would expect under purely opportunistic criminal targeting, suggesting a degree of geopolitical selectivity in victim choice.

The most attributed initiator, which is in line with intelligence community observations and operations of the well-known groups of Russian-speaking criminal gangs like REvil, Darkside, and Conti, was Russia. The most prevalent dyad of ransomware was Russia-to-United States, which aligns with reported geopolitical conflict. The total absolute

ransomware burden of nonstate groups is the highest, which is expected based on the financially incentivised RaaS model typified by [24].

The geopolitical environment and the data collection methodology may have contributed to the observed NATO-aligned concentration itself. The nature of EURepoC database, whereby it is focused on politically motivated incidents, biases towards incidents that have attained geopolitical salience. Ransomware attacks targeting NATO-aligned countries have a higher chance of attaining this salience due to the fact that they target organisations whose failure will provoke concerns among the population, media attention, and governmental intervention. This selection mechanism can introduce an upward bias in the measured NATO-aligned rate as compared to the actual population rate, and ought to be considered a confounding variable in the interpretation of the H2 results.

The CENTAM anomaly, with a ransomware rate of 30.4 based on just 23 cases, highlights the need for caution with regional statistics with a small sample. This wide binomial confidence interval (about 12 percent-54 percent) makes this figure consistent with the NORTHAM rate of the lower end and much higher rates of the upper end. Two structural hypotheses are possible: small CENTAM samples in the database might be overrepresented with high-profile incidents due to their reflective nature of only material incidences being reported and attributed; or CENTAM incidents in the data might be government targets due to the dual financialgeopolitical nature of ransomware reported by [23].

The cross-analytical validation of the alignment between the EDA initiator analysis that presents Russia as a leading attributed ransomware initiator and the feature importance analysis that presents `attribution_id` as the most significant predictor. A threat intelligence capability to keep up-to-date attribution information regarding the most active ransomware-related groups of actors may be used to deploy the entire model with maximum possible performance. On the other hand, in functions that do not have attribution information, but have access to metrics of incident severity, the severity-only model variant could be used, and it would achieve $AUC > 0.91$ at a cost of about 4 percentage points. The decision between these deployment configurations is an operational decision which the ablation results now offers a quantitative advice to inform. The observed concentration of ransomware incidents in NATO and North American regions may also be influenced by geopolitical reporting asymmetries. Western countries typically have more transparent reporting mechanisms and

stronger attribution capabilities, which increases the likelihood that incidents are recorded in datasets such as EURepoC. Consequently, while the regional bias is statistically significant, it should not be interpreted as a complete representation of global ransomware distribution.

6.3 Machine Learning Performance

The 15-percentage point difference between ROC-AUC and PR-AUC is used to measure the extent of overstating performance by ROC-AUC in the case of imbalanced classes. This gap has hardly been reported in similar published studies. This notwithstanding, PR-AUC=0.6380.787 affirms the fact that the models are truly useful in detecting ransomware attacks. This result extends the findings of [4] and [5], whose endpoint-based classifiers reported high ROCAUC values but did not report PR-AUC or dummy baselines, making direct comparison difficult; the present study suggests that such omissions may lead to an overstatement of practical utility in imbalanced ransomware classification settings.

Logistic Regression has the same recall as XGBoost (0.87) but produces 72% more false positives (181 vs 105). In a Security Operations Centre (SOC) environment, excessive false alarms lead to fatigue when it comes to alerts and wastage in analyst time. XGBoost is able to do the same recall with a richer non-linear decision surface, and at the same time achieve a higher precision. XGBoost is then the model that should be deployed because it has a high recall, but far more precise.

The analysis of the Logistic Regression versus XGBoost on ransomware recall (both having 0.87) and XGBoost has much higher AUC (0.9593 vs 0.9200) and precision (0.58 vs 0.44) is not devoid of analytical interest. Both models detect the same percentage of ransomware incidents at the default threshold of 0.5. Nevertheless, XGBoost accomplishes this by a more calibrated probability output, i.e. the probabilities it forecasts better represent true posterior probabilities. When a SOC operator would like to dynamically adjust the threshold with the level of threat, the better-calibrated outcomes of XGBoost would be more accurate than the aggressively reweighted probabilities of Logistic Regression. This operational difference is critical in organisations which tune their detection sensitivity depending on the prevailing threat intelligence.

It is a well-grounded choice to omit deep learning in the context of the given dataset but presents a possible performance limit. The insignificant difference between the results of the

Random Forest (0.9585) and the XGBoost (0.9593) implies that the two models are close to the performance that is possible based on the number of numeric features used. The near overlap between two architecturally different ensemble methods, with almost identical levels of performance, is an indication that the discriminating signal in the existing feature set has been saturated, and additional performance would require new types of features (categorical variables, textual descriptions), or entirely new modelling strategies that can learn the interactions between features that cannot be efficiently learned by tree-based splits.

The omission of deep learning is justified well in the present set of data but imposes a performance limit. The difference between Random Forest (0.9585) and XGBoost (0.9593) is not huge which implies that the models are close to the performance that could be achieved using the available numeric features. More flexible architectures could be useful in the future in the work that involves categorical features or textual descriptions of incidents.

6.4 Feature Importance and Deployment Concerns.

Eliminating `attribution_id` cut AUC by just 1.7 percentage points, indicating that the model continues to perform well without information on the who did the attack. The most important practical result is the severity-only ablation AUC 0.912-0.918: with no attribution data needed, a severity-based model variant has high operating utility. This finding aligns with [2], who demonstrated that strategic-level signals without technical attribution data retain meaningful predictive power for ransomware classification, suggesting that operational and severity features carry sufficient discriminative information for deployment contexts where attribution is unavailable. Security operations functions that do not have a real-time attribution infrastructure can implement this variant bearing the cost of a small performance reduction.

Start year feature is the most dominant SHAP value which indicates the actual increase in ransomware since 2015. But it also implies that once the ransomware rates even out in future, the model could become less accurate as time goes on and would require to be re-trained on fresh data. This concern of temporal validity suggests retraining at regular intervals when new releases of EURepoC are made. In practical deployment, attribution features may not be available at prediction time, limiting real-world applicability.

The biggest gap that has been left in this paper is the fact that the `receiver_regions` variable is not a part of the machine learning models. This was a statistically significant variable with ransomware (chi-square $p=1.04e-41$), but was not included due to its multi-value string format that is complex. The most significant step towards improving the models is encoding it via onehot encoding which has less than 30 categories.

6.5 Practical and Policy Implications

To threat intelligence practitioners, the results give them a strong prioritisation roadmap: events with high severity scores, targeting NATO-aligned targets, with attribution to known actors, and post 2015 events should be high-priority candidates to ransomware investigation, even prior to the availability of complete technical analysis.

To policy makers, the ransomware geographical concentration that is documented is an investment justification in collective cyber defence systems under the NATO and EU regime. The information gaps that currently restrict the analytical work would also be addressed by fixing the mandatory incident reporting and coordinated attribution processes. The temporal observation that ransomware has been growing significantly since 2015 is aligned with the nature of policy literature to describe ransomware as a growing strategic threat that must incur sustained policy actions as opposed to reactive ones [3].

In the case of Security Operations Centres (SOCs), XGBoost and Random Forest classifiers of the kind described in this study might be incorporated into SIEM platforms to automatically rank incoming incidents based on probability of ransomware. This would assist the analysts in concentrating their attention to the most probable ransomware cases without having to manually sift through all the alerts. Human analysts should not be replaced by machine learning classifiers but rather be assisted. An incident should be guided by a model output that shows high probability of ransomware but human judgement should be used to make final decisions.

6.6 Summary

The two hypotheses are both strongly and convergently supported. XGBoost has AUC 0.9593 and PR-AUC 0.787, much higher than baselines of dummy. Without the presence of attribution data, feature ablation confirms severity-based deployment pathway with

AUC>0.91. Improvement of the categorical feature exclusion and temporal validity area is the primary areas of future improvement.

A more general question that the study poses is the correct place of machine learning in strategic cybersecurity decisions. Machine learning classifiers do not generate binary decisions but instead generate probabilistic ones, estimates of the likelihood of a particular event being ransomware. These outputs are not to be used as a decision-making system, but as one of the inputs to a human analytical judgement process. An output of the classifier of 0.85 probability of ransomware should elevate the previous estimation made by an analyst and start to allocate more investigative resources to the incident, yet should not create a pre-programmed reaction without human involvement. This is not a constraint of the technology but a suitable constraint of high stakes security situations in which false positives and false negatives have an unequal cost which a general-purpose classifier cannot adequately predict. The precision-recall analysis of this research gives the analysts empirical calibration of the model outputs to weight them accordingly in their respective operational settings.

Chapter 7: Conclusion

7.1 Overview of the major results

The present dissertation examined recent ransomware attacks based on the EURPoC Dyadic Cyber Incident Dataset with the help of the supervised machine learning pipeline. The exploratory data analysis identified a statistically validated structural break in the prevalence of ransomware at 2015 with the prevalence increasing to 2.1% up to 16.3% ($p < 0.001$). The effects sizes were very large in all three measures of intensity (Cohen's $d = 1.23-1.31$, $p < 0.001$) which supports H1 and confirms that ransomware incidents are significantly more effective than other types of cyber incidents. H2 was supported by regional analysis which revealed disproportionate ransomware density in NATO-aligned and North American regions (chisquare $p = 1.04e-41$).

Ensemble model test results showed AUC of machine learning were over 0.958 in both models which is far above the 0.500 dummy baseline. The PR-AUC values of 0.638-0.787 verified excellent performance of the minority-class at the presence of imbalance between classes. The dependency on attribution was measured through feature ablation, which revealed a severity-only model could maintain $AUC > 0.91$ independent of attribution data, giving it a viable pathway to classification in real-life. SHAP analysis has validated that start year, attribution id, and severity metrics are the most predictive ones with convergent validation by two independent importance estimation procedures.

7.2 Achievement of Research Objectives

Five objectives were met. Extensive EDA was done in terms of temporal, regional, sectoral, and severity trends (Section 5.1, Figures F.1-F.22). Three classifiers under supervision were constructed and tested with class imbalance treatment (Section 5.2, Tables G.1-G.3). A complete set of metrics such as PR-AUC and dummy baselines were used as measures of performance (Tables 5.3-G.5). The feature importance and SHAP analysis were done by using feature ablation (Sections 5.2.6-5.2.7, Figures 5.6-F.33, Tables 5.4-G.4). Findings were related to existing literature throughout Chapter 6. Addressing the four research questions directly: Research Question 1, concerning major trends in ransomware occurrence, is addressed by the temporal and regional EDA in Sections 5.1.2 and 5.1.4, which identified the 2015 structural break and NATO-aligned concentration. Research Question 2, concerning the extent to which ML models can classify ransomware from incident-level features, is answered

through the model evaluation in Section 5.2.1, where ensemble models achieved AUC >0.958, far exceeding the dummy baseline. Research Question 3, on which characteristics are most significant, is answered through the SHAP and Gini importance analyses in Section 5.2.6, identifying `attribution_id`, severity metrics, and `start_year` as primary predictors. Research Question 4, on differences in severity and regional targeting, is addressed by the H1 severity analysis (Section 5.1.3, Cohen's $d=1.23-1.31$) and the H2 regional analysis (Section 5.1.4, $\chi^2 p=1.04e-41$).

7.3 Hypothesis Outcomes

H1 was found correct: ransomware has very large differences in severity between nonransomware incidents ($d= 1.23-1.31$, $p<0.001$ in three independent measures). H2 was validated: ransomware is disproportionately targeted at areas aligned with NATO ($p=1.04e41$) and over 20 times higher in some areas than others. Both results are strong, convergent in the various analytical procedures and aligned to the theoretical framework based on the literature review.

7.4 Original Contributions to Knowledge

This study contributes a machine learning framework for ransomware classification using the EURepoC Dyadic Cyber Incident Dataset, achieving a test AUC of 0.9593. To the author's knowledge, this study extends existing work by applying SHAP-based analysis to this dataset, identifying "start-year" as the primary discriminating feature and documenting elevated ransomware rates in the wastewater sector. Furthermore, the study validates a statistically significant severity gap between ransomware and other threats (Cohen's $d > 1.2$) and offers a deployment-optimised pathway (AUC > 0.91) refined via feature ablation. These results contribute to the literature on bridging geopolitical context with predictive modelling to inform defensive strategies.

7.5 Recommendations

As a security operations team: Use XGBoost or Random Forest classifiers to triage ransomware. XGBoost is the most preferred because they have a high recall (0.87) and a lower false positive rate (105 vs 181 with Logistic Regression). In the absence of attribution data, consider a severity-based model variant that does not lose AUC>0.91.

In the organisations located in NATO-aligned and North American areas: assume that a high level of ransomware is a structural, continuous issue that needs regular security investment, especially in healthcare, critical infrastructure, and government domains as highlighted in the sectoral review.

To policy makers: leverage the evidence of the regional risks to benefit collective cyber defence investment, coordinated attribution mechanisms, and obliged incident reporting in NATO and EU frameworks.

The Security Operations Centre tooling implications are action oriented. The XGBoost and Random Forest classifiers created in this paper can be incorporated into security information and event management (SIEM) systems as automated scoring policies to indicate incoming incidents as ransomware probable based on their recorded severity indicators, temporal context and attribution information when available. This automated pre-screening would ease the load on the analysts to manually review all incoming incidents, and focus on the most likely ransomware victims, whilst sending lower-probability incidents through regular triage. Since average SOC analysts look at hundreds of alerts every day, automated probabilistic prescreening is a feasible solution to enhance detection efficiency without necessarily adding extra headcount to the analyst teams.

Findings of the study can be added to a larger research agenda at the crossroads of computational social science, political science, and cybersecurity. The use of machine learning on geopolitically organized incident databases relates to trends in conflict studies and international relations, which have seen an increase in the use of quantitative approaches to structured data on political violence and diplomatic incidents. The methodological overlap of both approaches is high: both entail categorization of event types based on structured metadata, both need to deal with class imbalance and uncertainty in attribution and both are interested in interpretable results that can be taken into action by practitioners and policymakers. The interdisciplinary communication between these communities might benefit both disciplines, as machine learning methodological rigour may be applied to political science to analyse the incident and geopolitical theories to machine learning research in cybersecurity.

The interaction of the main limitations of the study introduces a cumulative uncertainty regarding the readiness to deploy that is bigger than the overall limitations. The selection bias

implies that the model might not extend to all ransomware; the categorical feature exclusion implies that the model makes use of the available signal; the temporal split concern implies that the performance is potentially optimistic; and the attribution dependency poses a deployment boundary. Both of them are controllable separately; they imply that the models are to be viewed as evidence of concept and not as a system to be deployed. The particular value, though, is setting out the methodological framework, and empirical baseline, not a production classifier, but a truthful characterisation which does not reduce the contribution of analysis but properly puts its applicability into proper context.

In spite of these cumulative limitations, the results of the study give a sound basis to two particular claims that are strong in all major constraints. To start with, the EDA results of ransomware having large severity differentials ($d > 1.2$) and being highly regionalised ($p = 1.04e41$) are true characterisations of the EURPoC dataset population whether that population is a true representative of actual ransomware. Second, the ML performance comparisons with AUC of above 0.958 versus 0.500 dummy baseline are valid in the evaluation framework employed, and the relative rankings of the models are not likely to flip under that of an alternative evaluation design even when there is change in absolute performance levels. These strong arguments form a significant contribution to the scientific field and the limitations merely qualify rather than nullify these arguments, which form the basis of evidence of the initial contributions made in Chapter 7.

The results of the study are also applicable to the policy of cybersecurity formulation on the national and alliance level. The measured regional distribution of ransomware threat, whereby NORTHAM and NATO-aligned territories shoulder a disproportional burden, forms a basis of proportionate investment in national cybersecurity and burden-sharing agreements. Policy discourses on ransomware currently are generally based on anecdotal reporting of high-profile attacks, and industry survey data of dubious quality. The dataset-level systematization offered in this research gives a more stringent base of policy calibration in that it does not only show that ransomware is a problem but quantifies its concentration, where it is intense in comparison with other hazards, and which attributes best forecast the ransomware.

In the case of cybersecurity training and education, the results prove that machine learning on strategic intelligence databases can be used to solve a separate category of security issues than endpoint detection. The existing cybersecurity curricula mainly view machine learning

as an endpoint anomaly detection and malware classification tool. This paper demonstrates that ML based on incident-level geopolitical data can solve strategic issues such as incident characterisation, threat prioritisation, and geopolitical risk assessment that demand varied types of data, feature engineering methods, and assessment models. The synthesis of threat intelligence domain expertise and the strictest methodology of ML give an example of the cross-disciplinary synthesis that this new field needs.

The methodological contribution of the study is on par with the empirical findings. The paper integrates stratified cross-validation, class reweighting, multi-metric analysis using dummy bases, SHAP interpretability with systematic feature ablation in one pipeline, contributing a methodological approach to how machine learning can be applied to structured datasets of cyber incidents.

The most straightforward operationally actionable finding is the feature ablation contribution. The empirical answer to the question of whether attribution data is useful to a model has a direct empirical answer: removing attribution features decreases AUC by 1.7 percentage points, and a severity-only model has $AUC > 0.91$. This gives security teams a straightforward deployment model: attribution-data teams should run the full model; attribution-data teams should run the severity-only model and optimize to the 4.7 percentage point AUC cost. This type of quantitative guidance, specific to deployment, is not found in most other similar published works, and is a methodological contribution that can be transferred across datasets as well as across this particular dataset.

The threat intelligence practice recommendations should have certain roots in the empirical results. Having identified the `start_year` as the strongest SHAP feature has a particular operational implication: any security analyst considering a newly-reported incident that took place after 2020 should place a greater prior probability of the incident being classified as ransomware than of the incident being classified as ransomware that happened before 2015. The severity metrics also give the particular threshold guidance: `unweighted_intensity` incidents of over about 3.5, about 1.2 standard deviations above the mean of non-ransomware, positively contribute to ransomware classification. These quantitative reference points, which are based on the SHAP analysis, give analysts empirically-based calibration aids in the manual ransomware risk assessment that are complementary to the automated classification pipeline.

It is worth noting the practical implication of the PR-AUC results. The imbalanced classification tasks commonly reported as ROC-AUC are in a systematic overstatement of model performance. ROC-AUC of 0.959 alone is a great number; PR-AUC of 0.787 on the same model is a more realistic description of its ability to detect ransomware. They are both sincerely good outcomes, and PR-AUC gives more actionable information to practitioners considering the usefulness of the model to address their detection goals. It is recommended that both metrics should be reported in future ransomware ML research.

The congruence of the EDA results with the ML results is a quality indicator of the analytical pipeline of particular importance. The severity differentials observed in the H1 EDA analysis are confirmed independently by the ML feature importance analysis, where the same severity metrics are discovered as one of the most important predictors. The feature importances of the `attribution_id` and the `dyadid` identify the regional concentration that is independent of the H2 EDA analysis. This cross-analytical convergence is much more convincing evidence than either the EDA or ML result alone would be, since it eliminates the possibility that either of the results is an artefact of the particular methodology applied to obtain it.

Chapter 8: Limitations and Future Research

8.1 Principal Limitations

There are five limitations to this study. To begin with, the EURepoC dataset comprises only of publicly announced and officially accredited incidents. According to research, reported cases are less than 20% of reported volumes of ransomware [16]. The models might not be as effective with unattributed offenses of less high-profile criminal groups. This means that performance metrics reported here are likely optimistic relative to what would be achieved on a fully representative population of ransomware activity, and the classifier should be treated as best suited to high-salience, formally attributed incident environments rather than general deployment.

Second, the `receiver_regions` variable was very strongly related to ransomware ($p=1.04e-41$), but it was not included in the machine learning models due to its complicated multi-value string format. The most significant future enhancement is to include it by one-hot encoding. Its exclusion means that the model cannot directly exploit the strongest regional signal in the dataset, which likely places an upper bound on achievable AUC lower than what would be possible with full feature coverage.

Third, the train-test split was a random mix of all time periods incidents. In actual implementation, it would always test on incidents in future. A temporal split (train on pre-2022 and test on 2022-2024) would provide a more realistic approximation of the model performance in the real world, especially since `start_year` is the largest SHAP contributor. The use of a random split may therefore produce slightly optimistic AUC estimates, as the model has seen future-period examples during training, which would not be available in a true prospective deployment scenario.

Fourth, the research merely categorizes events as ransomware or not. Operational security teams would find it more helpful to have a multi-class framework that also defines other types of attacks. The binary framing reduces the practical utility of the outputs for analysts who need to distinguish between ransomware, data theft, and disruption incidents in real-time triage, all of which require different response protocols.

A fifth weakness is the temporal validity: with the threat space changing, models trained using previous data will become less accurate over time. The fact that `start_year` has a SHAP

dominance in the study indicates the real growth tendency over time but also implies that a model fitted using post-2015 data might not be reliable in case the rates of ransomware level off or decrease. Any real deployment would require periodic retraining on new releases of EURepoC. In practice, this means that model performance should be monitored against incoming incident data after deployment, and a retraining trigger should be defined based on observed performance drift rather than a fixed calendar schedule. The model may be affected by temporal drift, as ransomware tactics and threat actor behaviour evolve beyond the dataset period (post-2024).

8.2 Future Research directions.

1. Add one-hot encoded categorical variables, particularly receiver regions, and initiator category and contrast the performance with the numeric-only baselines set in this paper.
2. Perform a temporal out-of-sample validation: Use pre-2022 incidents to train and use 2022-2024 data to perform a more realistic estimate of performance.
3. Train a multi-class attack type model to accurately predict Disruption, Data Theft, Hijacking and Ransomware as independent targets.
4. Develop an online learning system that trains model parameters after each release of new EURepoC.
5. Add natural language processing (NLP) to the text description fields of the EURepoC dataset to extract semantic characteristics as not represented in the numeric columns.

8.3 Concluding Remarks

Ransomware is among the cybersecurity issues of the day. It integrates financial, operational, and geopolitical aspects in a manner that demands strategic and technical actions. This dissertation has shown that, using structured data on geopolitical incidents and using supervised machine learning, one can obtain accurate, interpretable, and useful ransomware classifications, which are significantly better than any naive baseline method. This study offers researchers and practitioners an evidence-based structure of ransomware threat understanding and prediction by revealing the primary signatures of ransomware, such as high severity, targeting by specific profiles, attribution by NATO, and after 2015. The

presented analytical tools and results are a strong basis of future studies and add to the existing body of evidence about data-driven cybersecurity.

The necessity of strict, data-driven methods of analysis will only enhance as the ransomware threat environment becomes increasingly sophisticated and problematic in terms of the geopolitical context. The methodological basis created by this paper, which combines systematic geopolitical statistics with the ethically sound machine learning analysis, offers a baseline of that current analytical research and a framework that could be expanded by future scholars. The joint effect between reporting bias and attribution bias in the EURPoC dataset forms a compounded selection effect to distinguish analytically. Reporting bias: bias at this level occurs at the point of incident discovery and disclosure; attribution bias: bias at this next level involves even of the reported incidents only those who are formally attributed are included in the data with the identity of the actor. The intersection is the set of incidents that most effectively reflects both publicly significant and geopolitically salient incidents that received a formal attribution. Any trained model based on this data is best forecasts precisely this kind of incident, the high-profile formally attributed operations and worst forecasts most actual ransomware which do not attain public visibility or official attribution. This is a boundary that must be recognized in order to be responsible in deployment.

Regardless of the identified limitations, the results of the study are a solid basis of two particular claims that are strong despite all the primary limitations. To begin with, the EDA results that ransomware has large severity differentials and active regional concentration are valid characterisations of the EURPoC dataset population with or without the population itself being representative of all real-world ransomware. Second, the performance comparisons of the MLs that indicate $AUC > 0.958$ over 0.500 of the dummy baseline are valid in the evaluation framework applied, and the relative model ranks will not change substantially under other evaluation designs despite the possible change in absolute levels. These strong statements are a significant scientific contribution to which the restrictions are relevant, but not invalid.

It is worth highlighting the professionalisation of the ransomware ecosystem since it has essentially changed the risk environment. Current RaaS activities are well-structured criminal networks, with dedicated development teams to maintain the malware arsenal, affiliate recruiting and operational functions to screen and onboard new operators, negotiating teams to handle communication with victims and payment processing, and data leak site operations

to publish stolen information of non-paying victims. This operational complexity implies that events that are recorded in the EURepoC dataset are not accidental, but the result of planned, resource-heavy criminal actions. These operations are analytically tractable using machine learning because the targeting, timing and severity of attacks are predictable due to financial incentives and operational constraints; supervised classifiers can be trained to classify ransomware and other types of incidents based on these systematic properties.

The implications of the financial architecture of the RaaS model on the targeting patterns that are evident in the dataset are significant. The developers of RaaS affiliate operators usually assess them by the amount of ransom collected per operation, which generates a strong financial incentive to target high-value targets in rich countries with the ability to pay high ransom. This profit-maximisation logic can explain the following characteristics of the regional distribution of the dataset: the concentration of ransomware in NORTHAM and NATO-aligned geometries where potential ransom payments are highest, the low rate in CENTAS and MENA where the ability to pay ransom is less, and the high volume of attacks in healthcare and government sectors where disrupting operations generates strong incentives to pay ransom as The RaaS business model thus not only predicts the rise of ransomware with time but also that it will specifically be focussed in the most financially appealing target groups.

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Appendix A: EURepoC Dataset: Key Column Schema

This appendix summarises the key variables from the EURepoC Dyadic Cyber Incident Dataset (2025 release) used in this study. Only analytically relevant variables included in the machine learning pipeline are presented.

Table A.1: Key Dataset Variables (n = 4,296)

Column Name	Type	Role	Description
start_year	Integer	Feature	Year extracted from incident start date
Ransomware	Binary	Target	1 = ransomware, 0 = other attacks
weighted_intensity	Continuous	Feature	Weighted severity score
unweighted_intensity	Continuous	Feature	Raw severity score
impact_indicator_score	Continuous	Feature	Composite impact measure
attribution_id	Integer	Feature	Threat actor identifier
dyad_id	Integer	Feature	Initiator–target country pair
Disruption	Binary	Feature	Operational disruption indicator
Data theft	Binary	Feature	Data exfiltration occurred
Hijacking with Misuse	Binary	Feature	System misuse by attacker
affected_entities_value	Continuous	Feature	Scale of affected entities
economic_impact_value	Continuous	Feature	Financial damage (sparse)

Appendix B: Python Analysis Pipeline (Core Implementation)

The complete annotated Python analysis pipeline is available as a Google Colab notebook:

[https://colab.research.google.com/drive/18mljYV8DTC_Amb9iKlAwo-Km5I1ygvza?usp=sharing]

Libraries used: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, shap, scipy. All random seeds fixed at random_state=42.

Appendix C: Hyperparameter Optimisation Results

This appendix presents the hyperparameter search space and optimal configurations identified using GridSearchCV with five-fold stratified cross-validation. The optimisation objective was ROC-AUC.

Table C.1: Hyperparameter Search and Best Configurations

Model	Parameter	Values Tested	Best Value	Best CV AUC
Logistic Regression	C	0.01, 0.1, 1, 10	0.1	0.9264
Random Forest	n_estimators	300, 500	300	0.9541
Random Forest	max_depth	None, 10, 20	None	:
Random Forest	min_samples_split	2, 5	5	:
Random Forest	min_samples_leaf	1, 2	1	:
XGBoost	n_estimators	300, 500	300	0.9515
XGBoost	max_depth	4, 6	4	:
XGBoost	learning_rate	0.05, 0.1	0.05	:
XGBoost	subsample	0.8, 1.0	1.0	:

Appendix D: Classification Performance: Detailed Results

This appendix provides the full classification performance of all models on the held-out test set (n = 1,289). These results support the analysis presented in Chapter 5.

Table D.1: Model Performance Summary (Test Set)

Model	ROC-AUC	PR-AUC	Accuracy
Logistic Regression	0.920	0.638	0.843

Random Forest	0.959	0.772	0.921
XGBoost	0.959	0.787	0.900

Table D.2: Ransomware Class Performance

Model	Precision	Recall	F1-Score
Logistic Regression	0.445	0.863	0.588
Random Forest	0.665	0.708	0.686
XGBoost	0.580	0.863	0.694

Table D.3: Confusion Matrix Summary

Model	TN	FP	FN	TP
Logistic Regression	941	181	22	145
Random Forest	1062	60	48	119
XGBoost	1017	105	22	145

Appendix E: Correlation Analysis of Key Features

This appendix presents the Pearson correlation coefficients between selected numeric features and the binary ransomware target variable. Only the most relevant features are included to support interpretation in Chapter 5.

Table E.1: Key Feature Correlations with Ransomware

Feature	Pearson r	Direction	Interpretation
unweighted_intensity	+0.406	Positive	Strongest severity-related predictor
weighted_intensity	+0.383	Positive	Consistent severity signal

impact_indicator_score	+0.371	Positive	Reinforces severity relationship
Disruption	+0.297	Positive	Ransomware linked to operational disruption
attribution_id	+0.293	Positive	Reflects concentration among actor groups
start_year	+0.068	Positive	Indicates temporal growth trend
Data theft	−0.041	Negative	Weak inverse relationship
Hijacking with Misuse	−0.073	Negative	Mild negative association

Appendix F: Figures and Visualisations

This appendix contains all figures produced by the Python analysis pipeline. Figures are organised into three sections: Section F.1 (Exploratory Data Analysis, Figures F.1–F.20), Section F.2 (Machine Learning Model Evaluation, Figures F.21–F.31), and Section F.3 (Feature Importance and Interpretability, Figures F.32–F.33). Per the module submission guidelines, all figures are appended with clear labels and captions. Section F.1: Exploratory Data Analysis (Figures F.1–F.20)

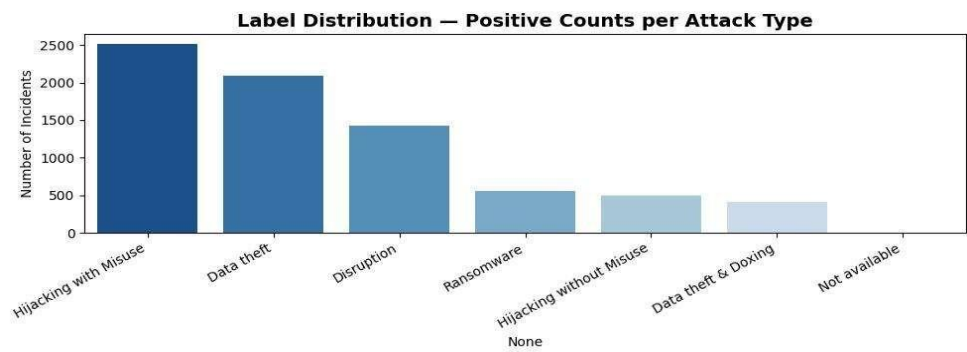


Figure F.1: Label Distribution: Positive counts per attack type (n=4,296). Hijacking with Misuse is the most frequent; Ransomware accounts for ~13.0%. Referenced in Section 5.1.1.

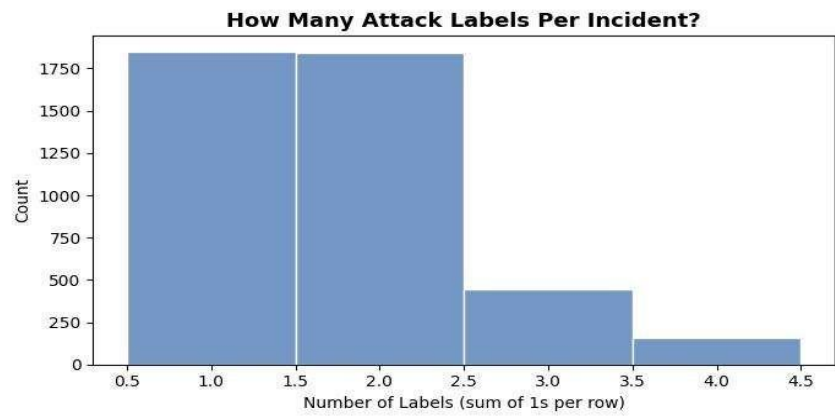


Figure F.2: Number of Attack Labels Per Incident: ~34% of incidents carry two or more labels, confirming the multi-label dataset structure (Section 5.1.1).

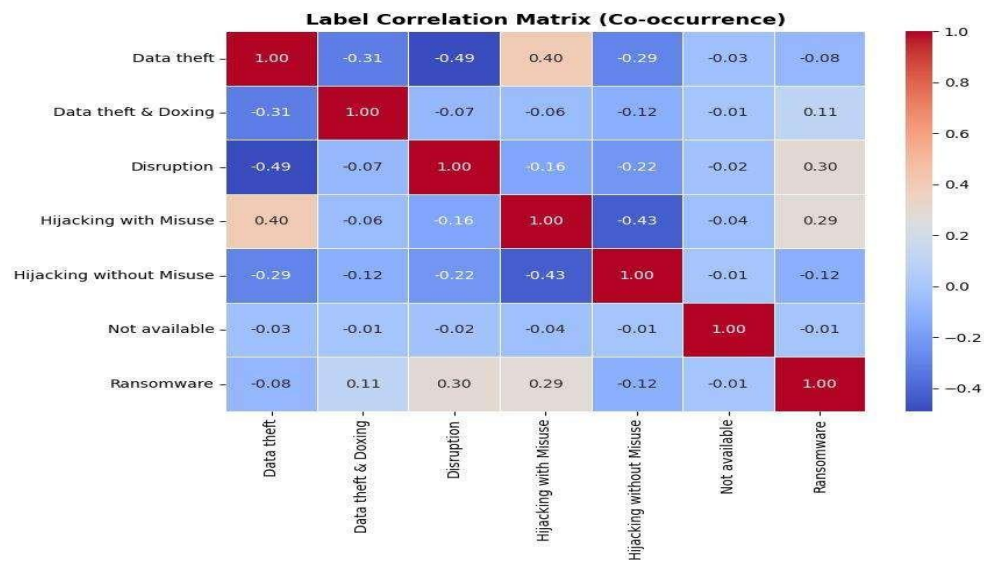


Figure F.3: Label Correlation Matrix : Ransomware shows positive co-occurrence with Disruption ($r=0.30$) and Hijacking with Misuse ($r=0.29$). Referenced in Section 5.1.1.

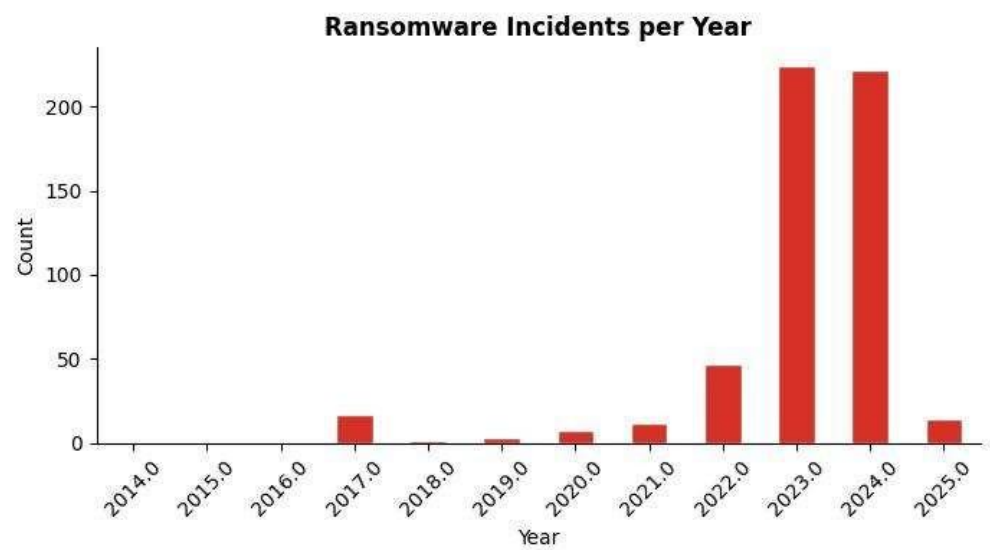


Figure F.4: Ransomware Incidents per Year: Clear upward trajectory from 2017, peaking in 2022–2024. Referenced in Section 5.1.2.

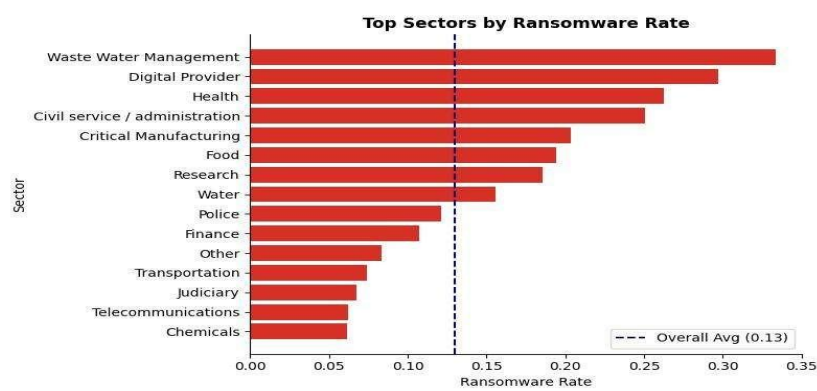


Figure F.5: Top Sectors by Ransomware Rate: Waste Water Management (33%), Digital Provider (30%), Health (26%) above dataset average of 13.0%. Section 5.1.5.

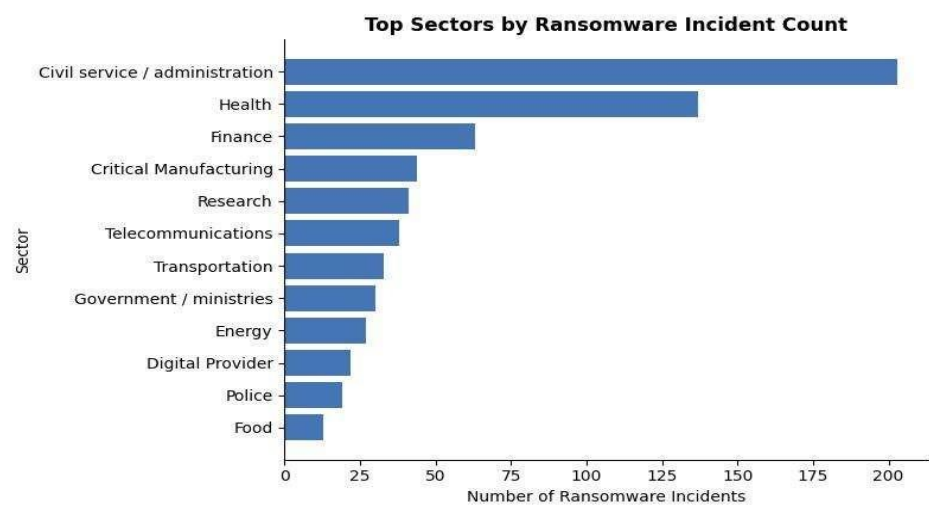


Figure F.6: Top Sectors by Ransomware Incident Count: Civil service / administration and Health account for the largest absolute burdens. Section 5.1.5.

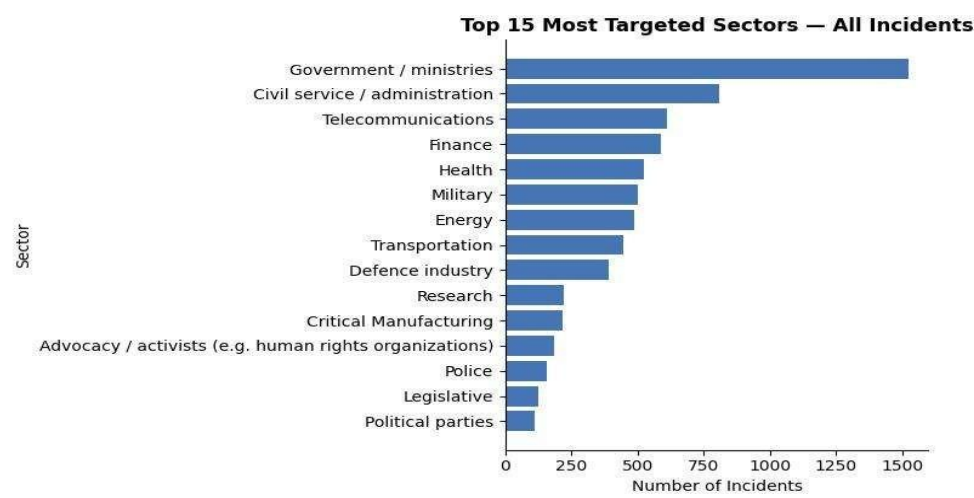


Figure F.7: Top 15 Most Targeted Sectors (All Incidents): Government / ministries dominates overall incident volume. Section 5.1.5.

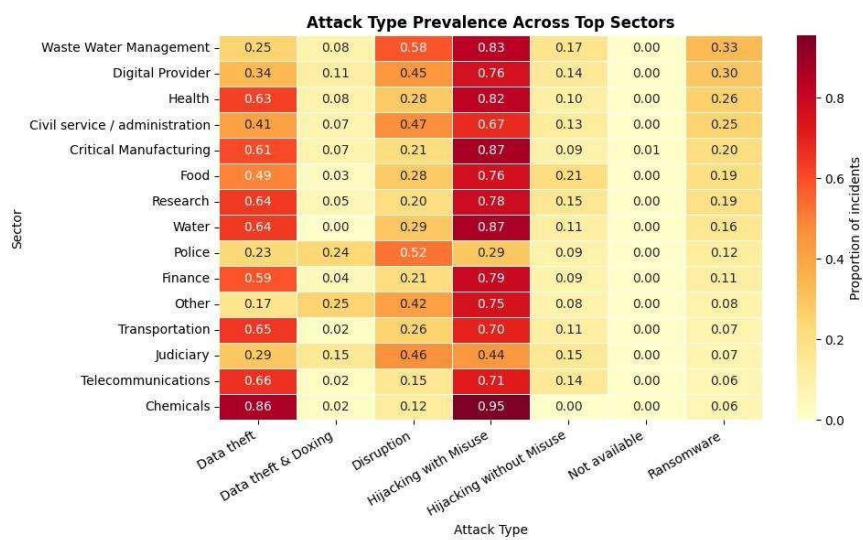


Figure F.8: Attack Type Prevalence Across Top Sectors: Ransomware concentrated in Waste Water, Digital Provider, and Health. Section 5.1.5.

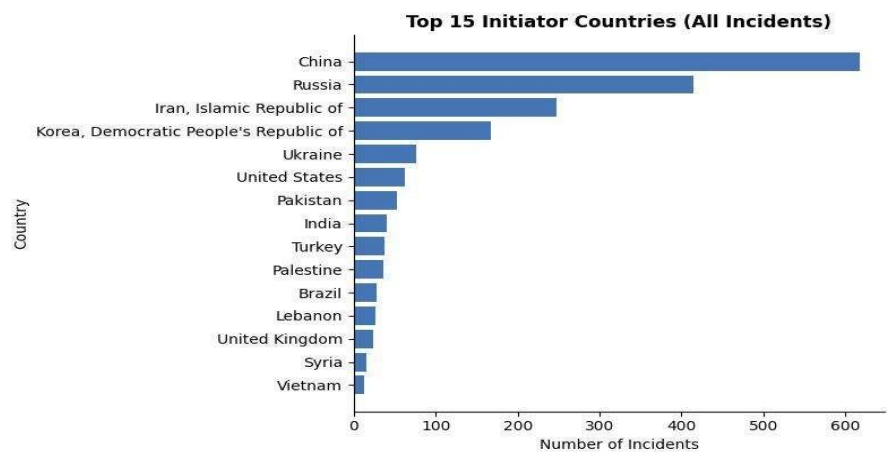


Figure F.9: Top 15 Initiator Countries (All Incidents) : China and Russia dominate attributed initiator counts. Section 5.1.5.

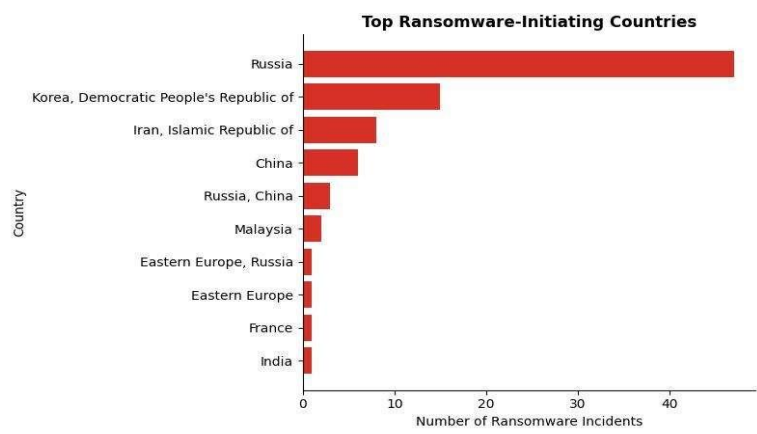


Figure F.10: Top Ransomware-Initiating Countries: Russia (~45) dominates attributed ransomware initiations; DPRK, Iran, and China follow. Section 5.1.5.

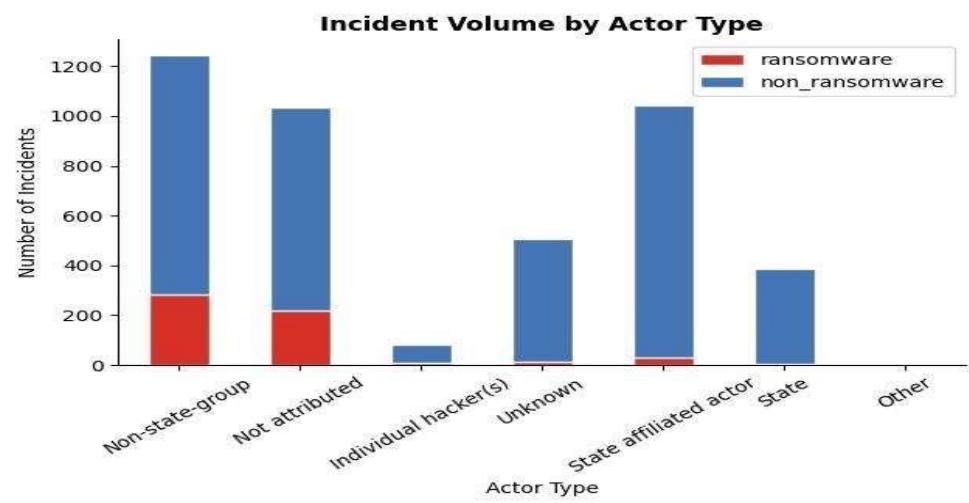


Figure F.11: Incident Volume by Actor Type: Non-state groups account for the largest absolute ransomware volume. Section 5.1.5.

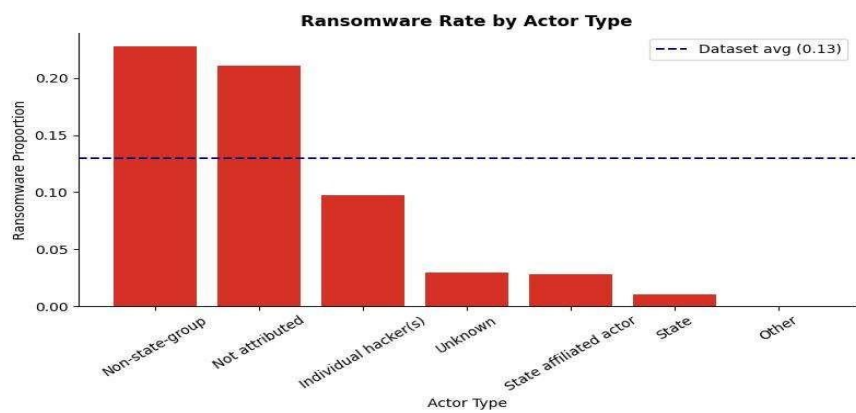


Figure F.12: Ransomware Rate by Actor Type : Non-state and unattributed actors show rates above the dataset average; state actors markedly lower. Section 5.1.5.

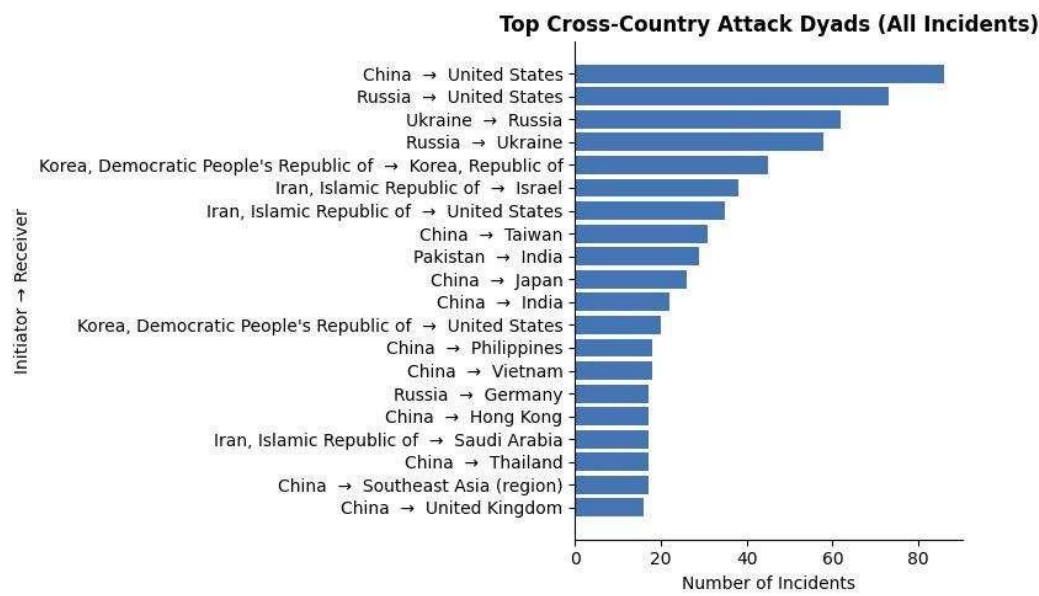


Figure F.13: Top Cross-Country Attack Dyads (All Incidents) : China→US and Russia→US are the two dominant dyads. Section 5.1.5.

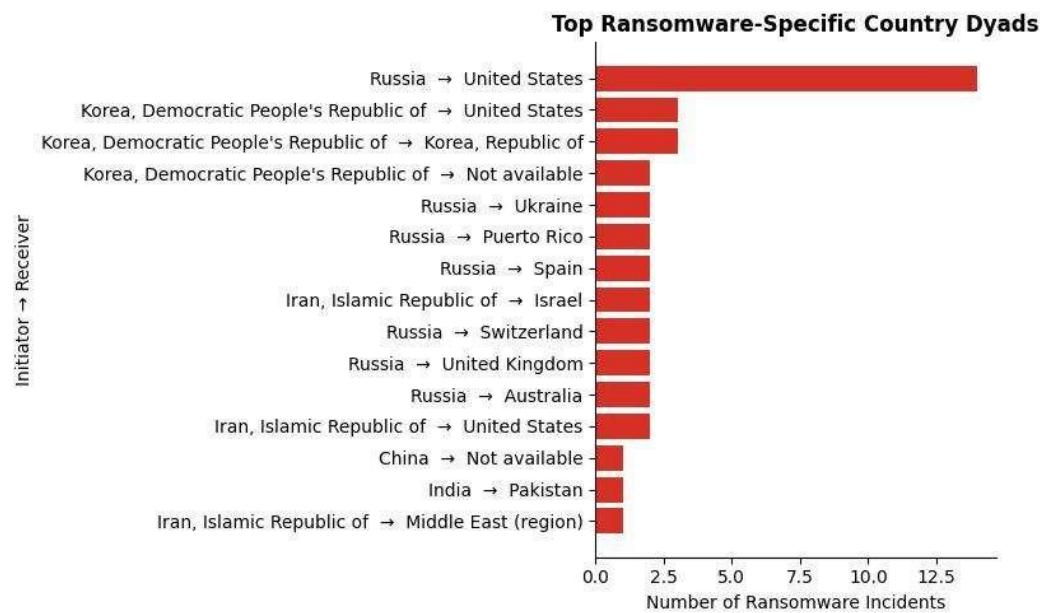


Figure F.14: Top Ransomware-Specific Country Dyads: Russia→United States ($n \approx 13.0\%$) is the dominant ransomware dyad. Sections 5.1.5 and 5.2.

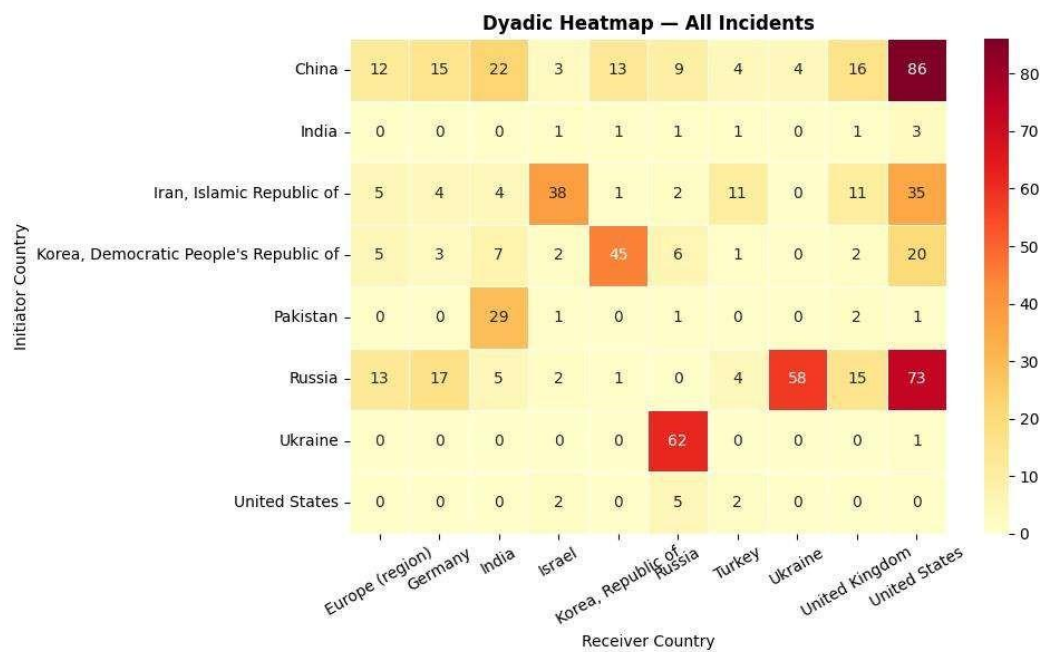


Figure F.15: Dyadic Interaction Heatmap: All Incidents: Cross-tabulation of incident counts between top initiator and receiver countries. Section 5.1.5.

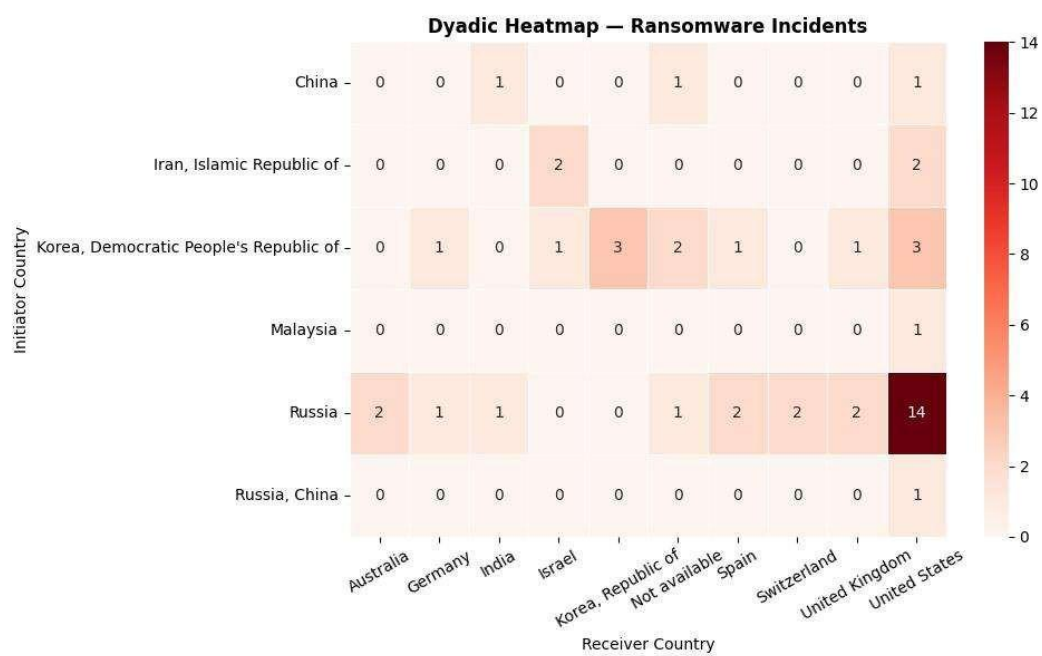


Figure F.16: Dyadic Interaction Heatmap: Ransomware Only: Russia→United States (14 incidents) is the dominant ransomware dyad. Sections 5.1.5 and 5.2.

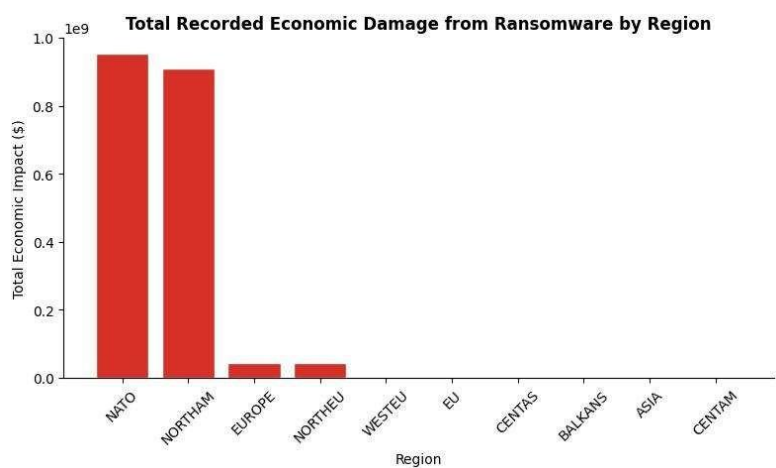


Figure F.17: Total Recorded Economic Damage by Region: NATO and NORTHAM dominate recorded financial losses from ransomware. Sections 5.1.5 and 5.1.4.

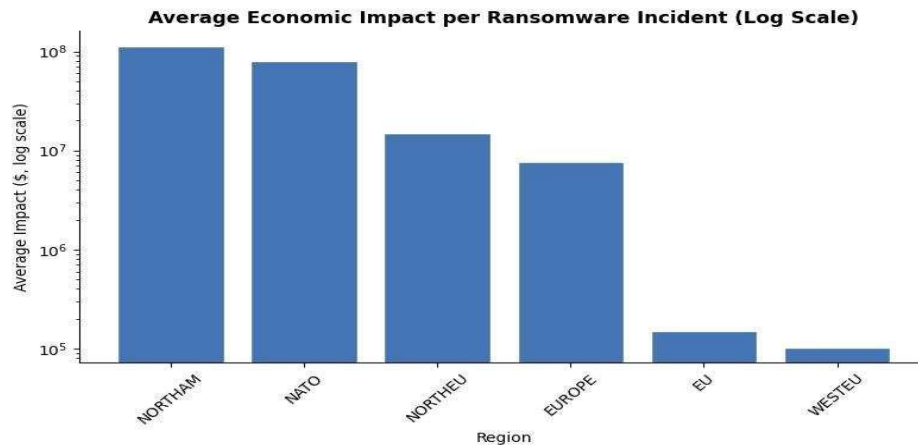


Figure F.18: Average Economic Impact per Ransomware Incident (Log Scale) : NORTHAM exceeds \$100M per incident average. Section 5.1.5.

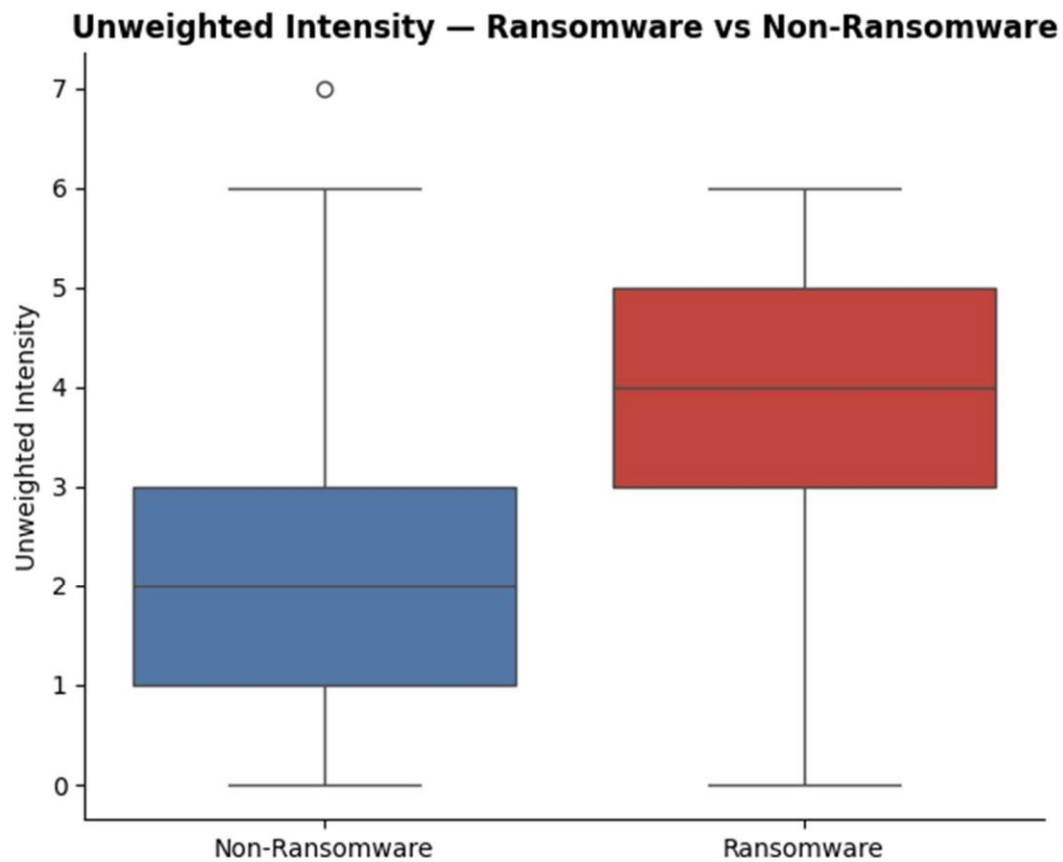


Figure F.19: Unweighted Intensity: Ransomware vs. Non-Ransomware: Box plot comparing `unweighted_intensity` distributions. Ransomware incidents show higher median values confirming the severity differential. Cohen's $d = 1.31$, $p < 0.001$. Section 5.1.3.

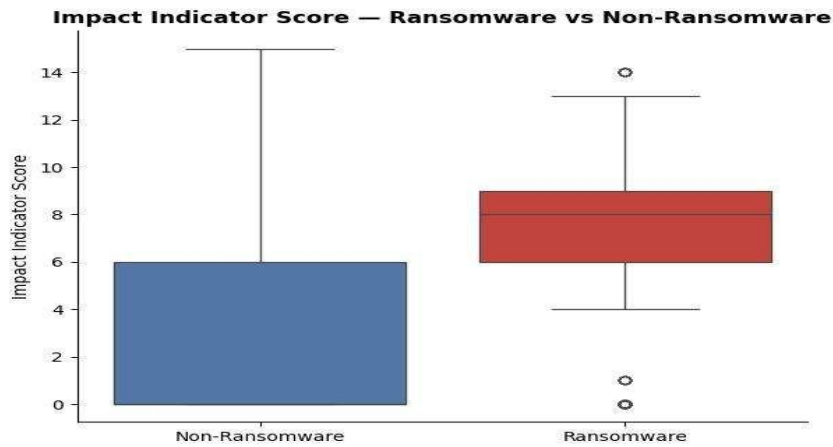


Figure F.20: Impact Indicator Score: Ransomware vs. Non-Ransomware: Box plot comparing *impact_indicator_score* distributions. Ransomware incidents show elevated median (7.8 vs 2.5 for non-ransomware). Cohen's $d = 1.28$, $p < 0.001$. Section 5.1.3.

Section F.2: Machine Learning Model Evaluation (Figures F.21–F.31)

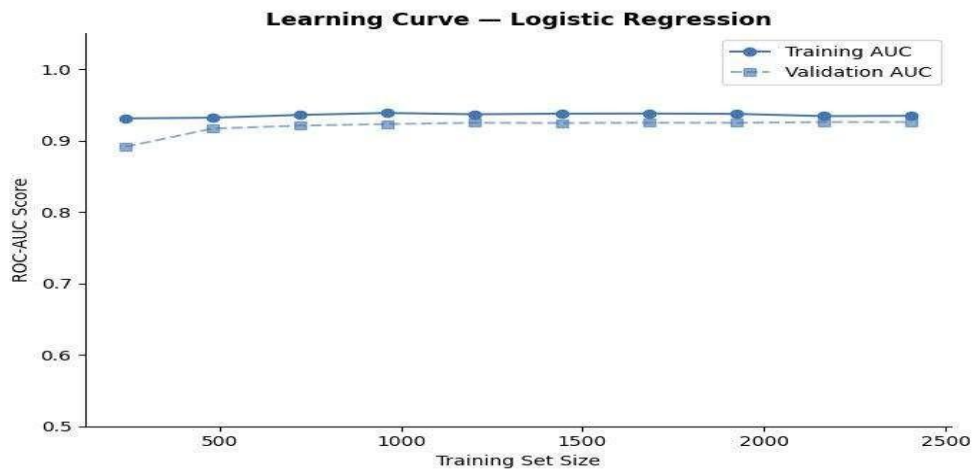


Figure F.21: Learning Curve: Logistic Regression: Training and validation AUC plateau and converge (~ 0.04 gap); no meaningful overfitting. Validation AUC ≈ 0.93 . Section 5.2.4.

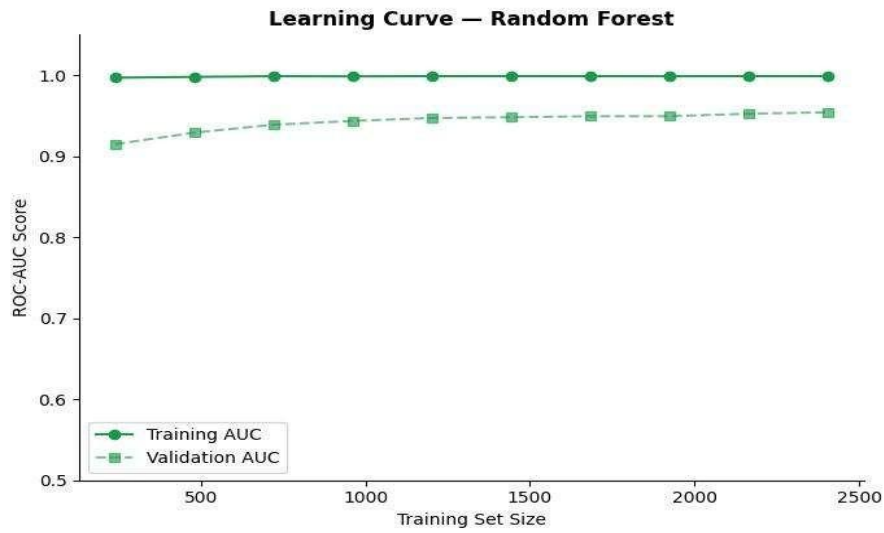


Figure F.22: Learning Curve: Random Forest: Validation AUC stabilises near 0.955; training-validation gap ≈ 0.045 . Section 5.2.4.

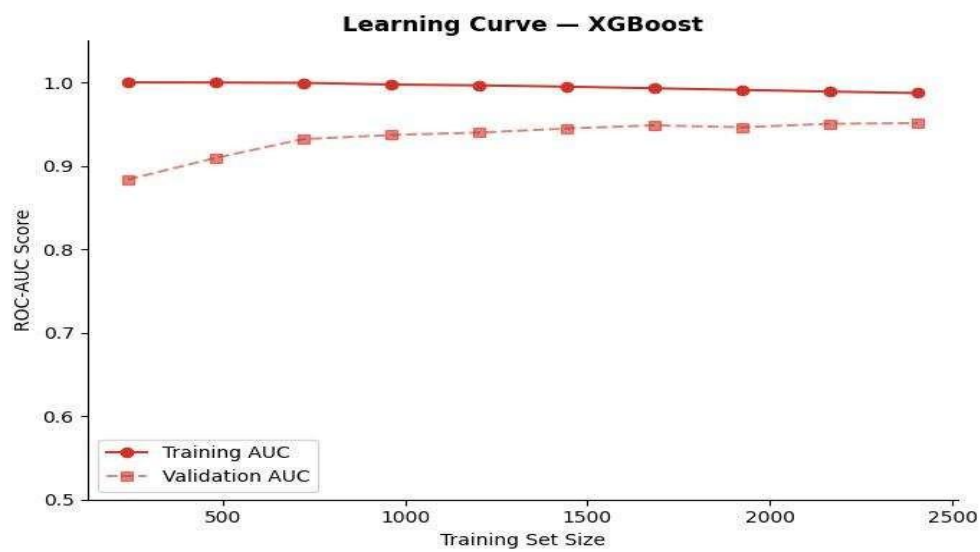


Figure F.23: Learning Curve: XGBoost: Validation AUC stabilises near 0.955; training AUC approaches 1.0 (normal for boosted ensembles). Section 5.2.4.

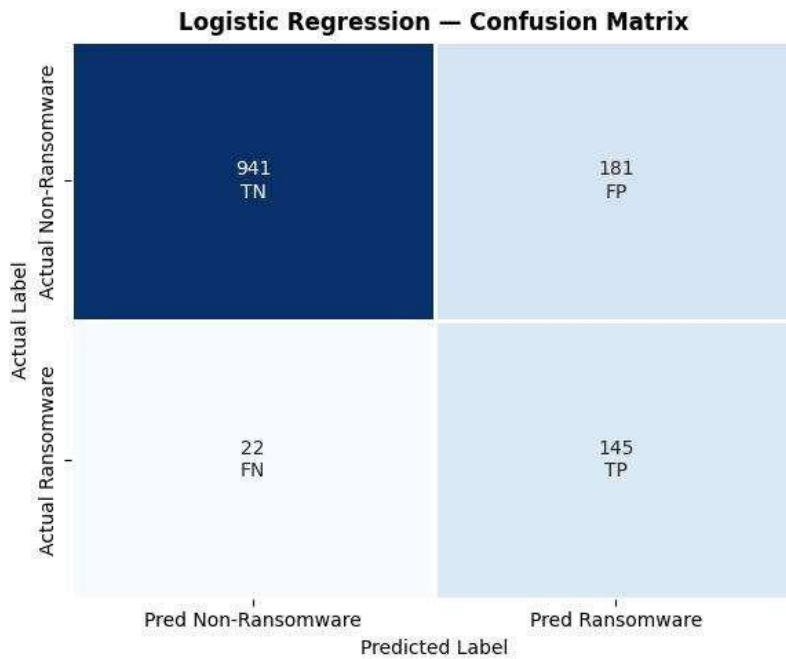


Figure F.24: Confusion Matrix: Logistic Regression: $TN=941$, $FP=181$, $FN=22$, $TP=145$. High recall (0.87) at cost of substantial false positives. Section 5.2.2.

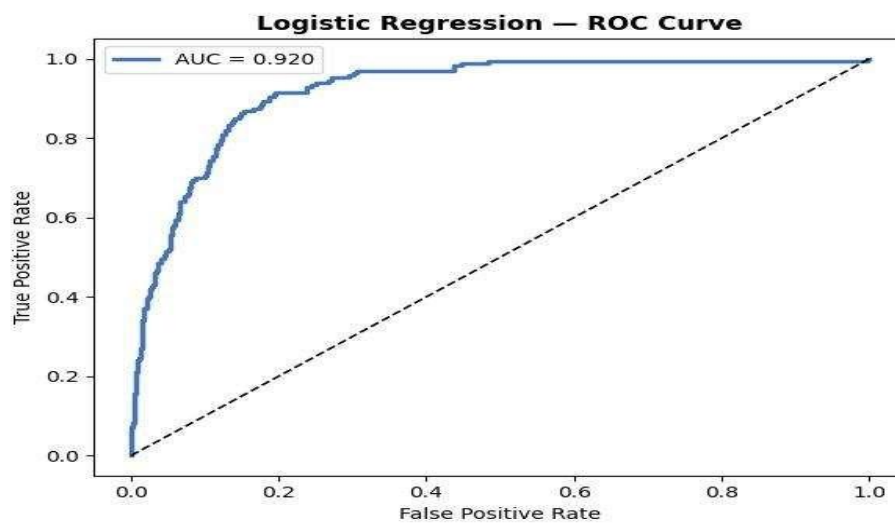


Figure F.25: ROC Curve: Logistic Regression ($AUC=0.920$). Section 5.2.2.

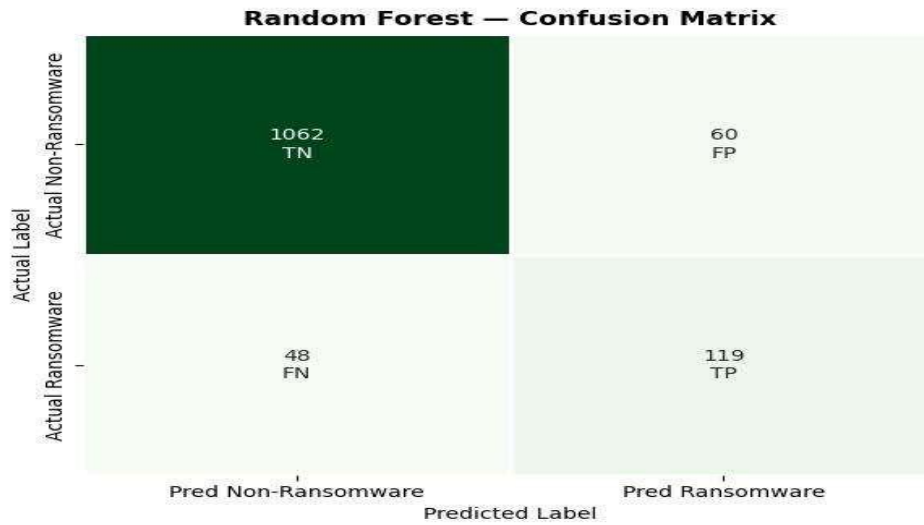


Figure F.26: Confusion Matrix: Random Forest: $TN=1062$, $FP=60$, $FN=48$, $TP=119$. Most balanced error distribution. Section 5.2.2.

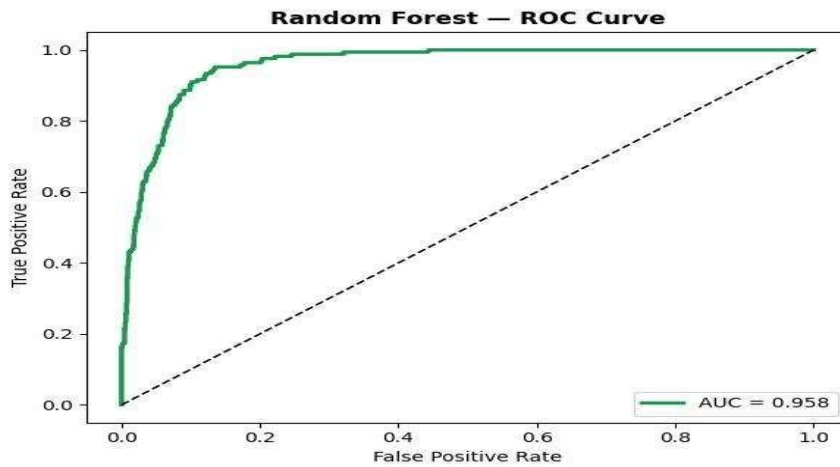


Figure F.27: ROC Curve: Random Forest ($AUC=0.958$). Section 5.2.2.

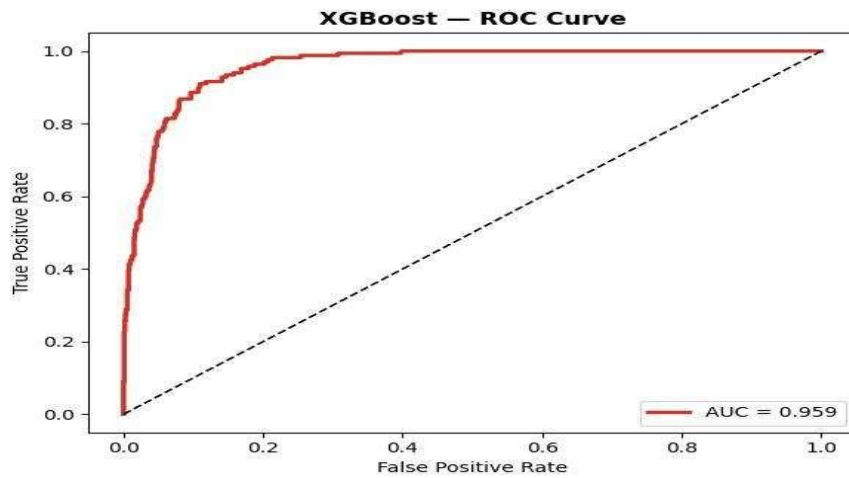


Figure F.28: ROC Curve: XGBoost (AUC=0.959), highest of all classifiers. Section 5.2.2.

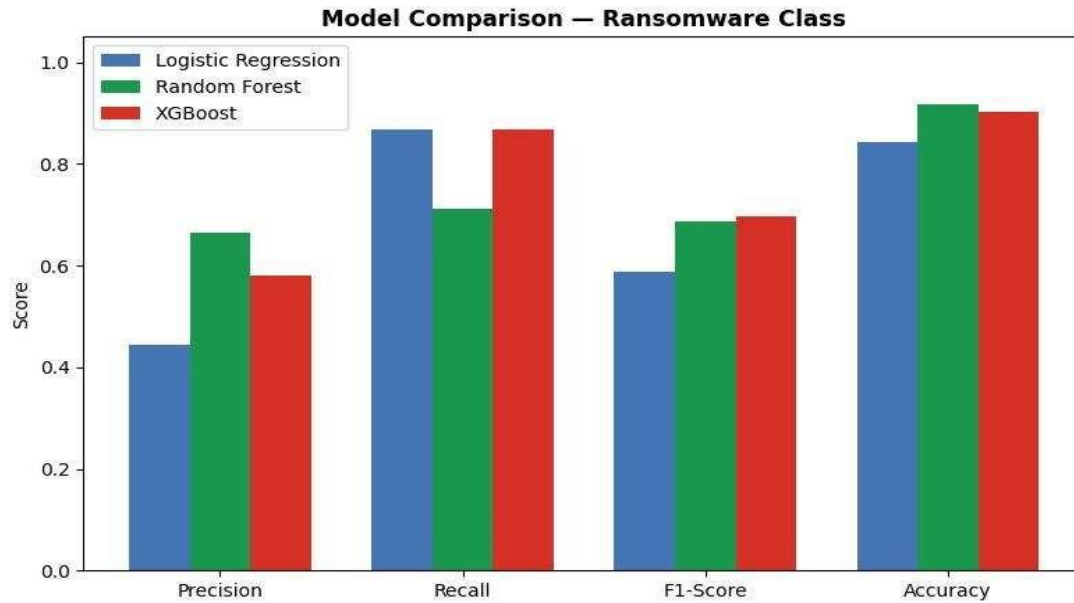


Figure F.29: Model Performance Comparison: Ransomware Class: XGBoost best F1 (0.70); LR and XGB tie on recall (0.87); RF highest precision (0.66). Section 5.2.2.

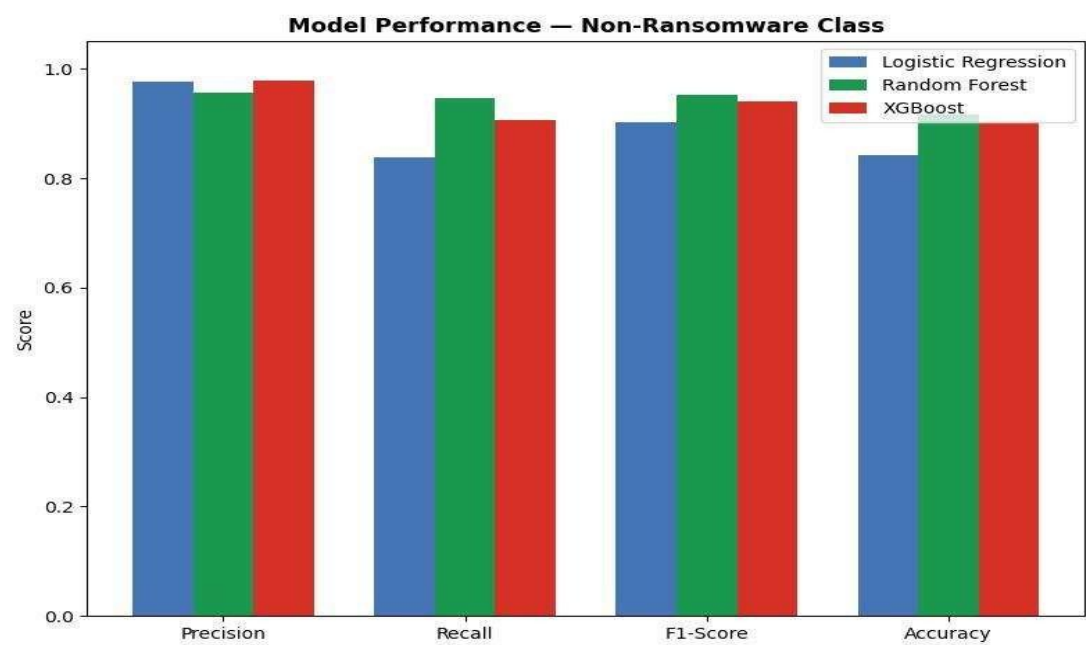


Figure F.30: Model Performance Comparison: Non-Ransomware Class: All models achieve precision and F1 above 0.90 for the majority class. Section 5.2.2.

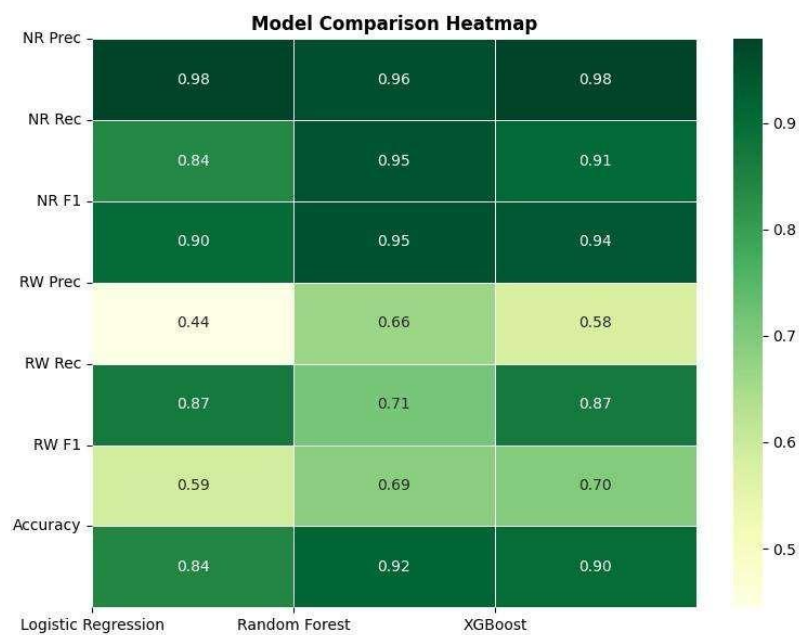


Figure F.31: Model Comparison Heatmap: Summary of all key metrics across all models for both classes. Section 5.2.2. Section F.3: SHAP Interpretability

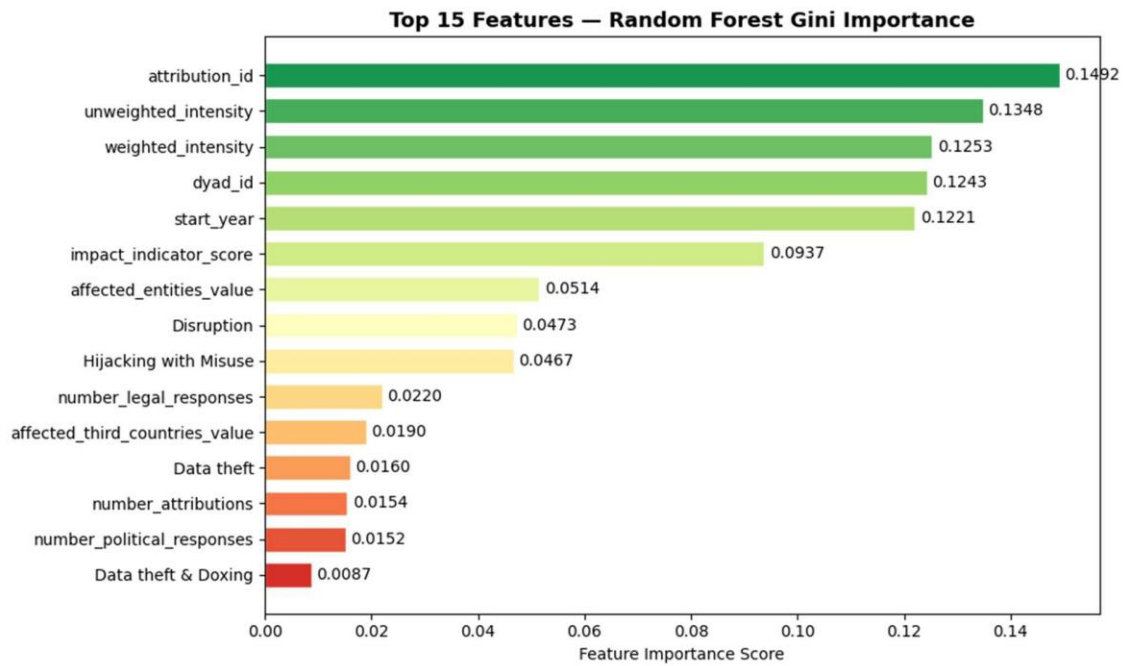


Figure F.32: Feature Importance: Random Forest: Top 5 features account for 65.5% of Gini importance. Section 5.2.6.

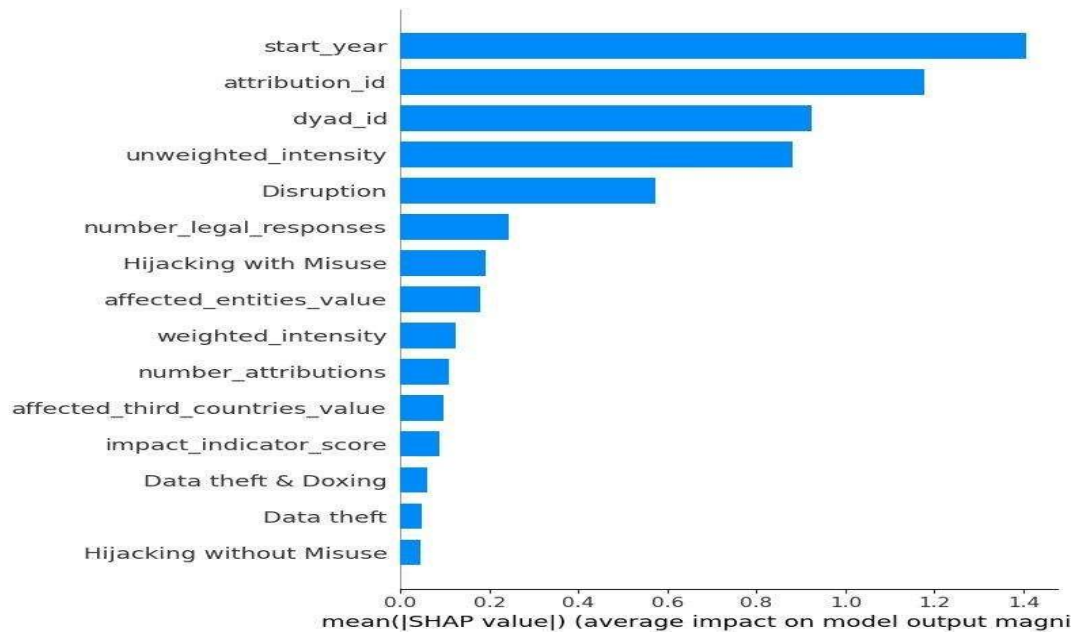


Figure F.33: SHAP Summary Plot: XGBoost: Beeswarm plot showing feature contributions to ransomware prediction; high severity scores and later start years push predictions towards ransomware. Section 5.2.6.

Appendix G: Summary Tables

This appendix provides supporting classification reports, feature importance rankings, and dummy baseline comparisons referenced in the main analysis.

Table G.1: Classification Report : Logistic Regression

Class	Precision	Recall	F1-Score	Support
Non-Ransomware	0.98	0.84	0.90	1,121
Ransomware	0.44	0.86	0.59	168
Weighted Avg	0.91	0.84	0.86	1,289

Table G.2: Classification Report : Random Forest

Class	Precision	Recall	F1-Score	Support
Non-Ransomware	0.96	0.95	0.95	1,121
Ransomware	0.66	0.71	0.69	168
Weighted Avg	0.92	0.92	0.92	1,289

Table G.3: Classification Report : XGBoost

Class	Precision	Recall	F1-Score	Support
Non-Ransomware	0.98	0.91	0.94	1,121
Ransomware	0.58	0.86	0.69	168
Weighted Avg	0.93	0.90	0.91	1,289

Table G.4: Gini Feature Importance Rankings : Random Forest

Rank	Feature	Gini Importance
1	attribution_id	0.149

2	unweighted_intensity	0.135
3	weighted_intensity	0.125
4	dyad_id	0.124
5	start_year	0.122
6	impact_indicator_score	0.094
7	affected_entities_value	0.068
8	economic_impact_value	0.054
9	Disruption	0.048
10	affected_third_countries_value	0.041

Table G.5: Dummy Classifier Baseline Comparison

Classifier	AUC	Accuracy	RW Recall
Most Frequent Dummy	0.500	0.870	0.000
Stratified Dummy	0.500	0.762	0.131
Logistic Regression	0.920	0.843	0.863
Random Forest	0.959	0.921	0.708
XGBoost	0.959	0.900	0.863